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Design and Fabrication of a Reliable Dual Recovery System for the N4 Rocket with Extended Telemetry Range

(FYP22-4)

Project Report

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June 9, 2026

Declaration

We declare that this final report is our original work and has not been presented for examination in any other institution. We confirm that all sources of information, data, and published material used in preparing this report have been appropriately acknowledged and referenced in accordance with university academic standards. We further affirm that this report reflects the complete work carried out in our Final Year Project and is a true account of all activities completed.

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Glenn Gatiba and Ian Joseph

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Abstract

The Nakuja N-4 rocket recovery system addresses three critical mechatronic failure modes documented in preceding missions: insufficient telemetry communication range, uncontrolled nichrome-wire igniter burnout during pyrotechnic parachute deployment, and an absence of real-time pre-flight electrical status feedback. This report details the complete design, fabrication, validation, and field testing of the mechatronic recovery suite developed to resolve these systemic challenges.

The telemetry subsystem utilises a Digi XBee-PRO 900HP transceiver operating within the 900 MHz ISM band under transparent Universal Asynchronous Receiver-Transmitter (UART) protocol. A first-order link-budget analysis validates a robust link margin of 32.34 dB over a 5 km line-of-sight recovery baseline. Field range testing confirmed sustained bidirectional packet delivery at distances exceeding 5 km, with battery voltage, altitude, and RSSI telemetry verified live on the ground-station liquid-crystal display (LCD).

The deployment ignition subsystem incorporates low-side switching driven by logic-level IRLZ44N Power MOSFETs regulated via a calibrated 70% Pulse-Width Modulation (PWM) scheme. Dual deployment pop tests confirmed successful drogue and main parachute ejection without nichrome wire burnout under repeated firings. System diagnostic feedback is provided via a 16-bit ADS1115 Analog-to-Digital Converter (ADC) over Inter-Integrated Circuit (I2C), supplying ground operators with real-time loop continuity verification and battery bus voltage tracking with a reconstructed accuracy of ± 0.02 V.

Mechanically, an additive-manufactured stepped Polyethylene Terephthalate Glycol (PETG) avionics bay sled secures four parallel 18650 lithium-ion cells within the 95 mm inner airframe diameter. A drone-mounted simulated-flight test confirmed active telemetry and correct FSM state sequencing. All mechatronic subsystems have been fabricated, integrated, and field-validated, establishing a

reproducible high-power rocket recovery standard for the Nakuja programme.

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Nomenclature

Abbreviations

ADC Analog-to-Digital Converter

AGL Above Ground Level

BMS Battery Management System

CAD Computer-Aided Design

ESP32 Espressif 32-bit Microcontroller Platform

FEA Finite Element Analysis

GPS Global Positioning System

IMU Inertial Measurement Unit

LiPo Lithium Polymer Battery

MOSFET Metal-Oxide-Semiconductor Field-Effect Transistor

PETG Polyethylene Terephthalate Glycol

PWM Pulse Width Modulation

RF Radio Frequency

UART Universal Asynchronous Receiver-Transmitter

Symbols and Units

D Duty cycle, percent

f Frequency, hertz

I Current, ampere

P Power, watt

R Resistance, ohm

T Temperature, degree Celsius

V Voltage, volt

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1 Introduction

1.1 Background of Study

Rocket recovery systems represent the most critical subsystem for preserving flight data and structural hardware in high-power rocketry missions. The evolution of recovery systems has progressed from simple single-deployment methods to sophisticated dual-deployment architectures. Single-deployment systems release a parachute at apogee, presenting risks of a high drift radius and unpredictable landing zones. In contrast, dual-deployment systems utilize a drogue parachute at apogee for stability, followed by a main parachute deployment at a lower altitude for a controlled soft landing, establishing the standard for modern high-power rocketry operations [1, 2].

The Nakuja Project is a student-led rocketry programme based at Jomo Kenyatta University of Agriculture and Technology (JKUAT) and the Pan African University Institute for Basic Sciences Technology and Innovation (PAUSTI), operating in collaboration with the Kenya Space Agency (KSA) and the Japan International Cooperation Agency (JICA) with the long-term goal of developing a liquid-propellant rocket capable of orbital payload delivery [3]. The programme has progressed through four rocket generations: N-1 (2021, simulated apogee 285 m), N-2 (2022, 500 m), N-3 (2024, 1,000 m target), and N-3.5 (May 2024, highest recorded apogee in the series at the Isiolo School of Infantry Range) [3]. Each generation has introduced lessons that inform the next design cycle.

The N-3.5 mission employed a piston-based single-deployment system using black powder and nichrome wire ignition, with telemetry transmitted over a Wi-Fi/MQTT link via a Ubiquiti NanoStation [?]. While the launch was successful, post-flight analysis documented thermal damage to the charge holder, nichrome wire burnout after repeated ground deployment tests, and an effective telemetry range limited to the Wi-Fi footprint of the NanoStation—constraints that

were insufficient for the higher-altitude, longer-range N-4 mission profile [?, 10]. The N-4 rocket, targeting an apogee of 3,540 m as an entry into the Spaceport America Cup (SRAD/Solid/10,000 ft class), requires a recovery system capable of operating at significantly greater ranges and with improved deployment reliability [3].

1.2 Problem Statement

The N-4 mission recovery system faces three documented critical challenges arising from N-3.5 field experience and the N-4 mission requirements:

1. **Telemetry range limitation:** The N-3.5 Wi-Fi/MQTT-based system (Ubiquiti NanoStation) was constrained to line-of-sight Wi-Fi range, which is insufficient for the N-4 simulated apogee of 3,540 m and competition recovery distance requirements exceeding 5 km. Signal degradation during critical descent phases risks total telemetry loss and an inability to locate the vehicle [?].
2. **Deployment unreliability:** Repeated N-3.5 ground tests documented nichrome wire burnout, thermal damage to the charge holder exceeding acceptable material limits, and inconsistent ejection from uncontrolled ignition energy. A single-deployment piston architecture also provides no redundancy against deployment failure at apogee [10].
3. **Absence of pre-flight system monitoring:** The N-3.5 system lacked real-time battery voltage and igniter continuity telemetry, preventing ground operators from verifying system readiness before launch and introducing the risk of a failed deployment from an undetected low-battery or open-circuit condition [?].

These three gaps collectively threaten mission success through potential vehicle loss, deployment failure, and unverifiable launch readiness. The proposed N-4

XBee telemetry architecture designed to address the first gap is illustrated in Appendix F (Figure 45).

1.3 Objectives

1.3.1 Main Objective

To design and fabricate a reliable dual-deployment recovery system for the N-4 rocket with an extended telemetry range.

1.3.2 Specific Objectives

1. To extend the telemetry range of the recovery system beyond 5 km using an XBee-PRO 900HP radio link.
2. To improve deployment reliability through controlled pulse-width modulation (PWM) pyrotechnic ignition, targeting repeatable nichrome-wire energization without wire burnout.
3. To design a rugged 3D-printed PETG avionics holder with vibration isolation, battery retention, and real-time ADS1115-based system diagnostics.
4. To validate the complete recovery system through ground ejection tests and field range tests.

1.4 Justification

The development of a reliable recovery system is paramount to mission success. Loss of the rocket vehicle results not only in hardware destruction but also in the complete loss of flight data critical for mission analysis and future design improvements. The N-4 rocket represents the Nakuja Project's most ambitious mission to date—a competition-class vehicle targeting 3,540 m—and the gap between its performance requirements and what the N-3.5 recovery system could

support is substantial. The proposed improvements address documented failure modes from N-3.5 field tests and implement dual-deployment and long-range radio practices that are standard in the high-power rocketry community.

Beyond the immediate mission, the open-source, student-centred model maintained by the Nakuja Project ensures that validated recovery designs and documented lessons become resources for future engineering cohorts and for the broader African student rocketry community [3].

1.5 Expected Outcomes

The expected outcomes of this study are:

1. Telemetry range exceeding 5 km with reliable XBee-PRO 900HP link performance demonstrated during range tests.
2. Consistent and repeatable parachute deployment through PWM-controlled pyrotechnic ignition without nichrome wire burnout.
3. A fabricated 3D-printed PETG avionics holder providing vibration isolation, battery retention, and real-time ADS1115 system diagnostics.
4. A validated dual-deployment recovery system demonstrated through ground ejection tests and integrated field testing.

2 Literature Review

2.1 Recovery Systems in Student Rocketry

Recovery systems preserve flight hardware and flight data and are therefore mission-critical in high-power student rocketry [1, 2]. A common practice is dual deployment: a drogue parachute at apogee for stability and reduced drift, then a main parachute at lower altitude for lower landing energy. Literature consistently shows that this staged strategy improves recovery reliability compared with single-deployment approaches in high-altitude missions [2].

The Nakuja Project’s own mission progression illustrates this trend directly. N-1 (2021) and N-2 (2022) used basic single-deployment parachute ejection. N-3 (January 2024) introduced on-board flight logging and improved parachute reliability but retained a single-deployment architecture. N-3.5 (May 2024) advanced to a piston-based pyrotechnic ejection system but still deployed only a main parachute, with telemetry relying on a WiFi/MQTT link via a Ubiquiti NanoStation [?, 3]. The N-3.5 post-flight analysis documented thermal damage to the charge holder, nichrome wire burnout after repeated ground pop tests, and effective telemetry range constrained to WiFi coverage—all of which define requirements for the N-4 system [?, 10]. Figures 1 and 2 show the N-3 dual-deployment context.

Gap 1 — Dual-deployment adoption: Prior Nakuja missions did not implement staged dual deployment. N-3.5 documented that single-deployment piston ejection provides no redundancy at apogee, and that the absence of a drogue parachute increases drift and descent-rate uncertainty at high altitudes. No mission-specific data linking charge mass, ejection effectiveness, and dual-deployment sequencing for the N-4 airframe geometry exists in the Nakuja record.

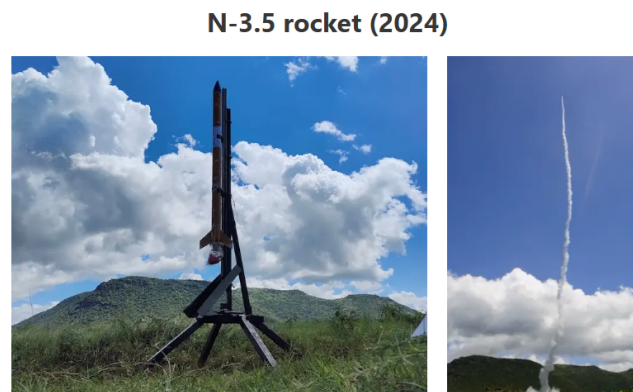


Figure 1: Nakuja N3 rocket with dual-deployment recovery configuration [3]

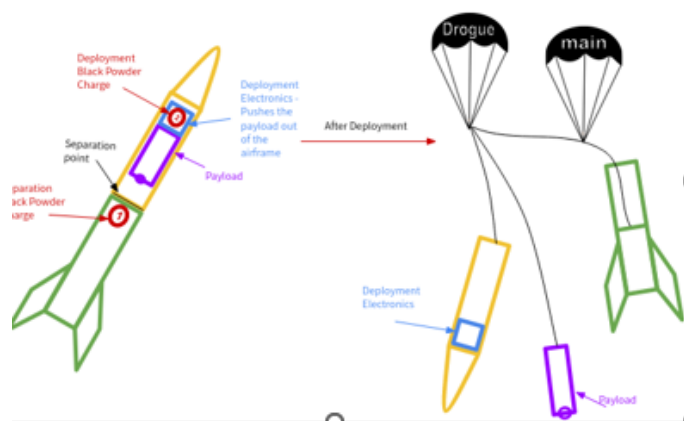


Figure 2: Nakuja dual-deployment flight during ascent [3]

2.2 Telemetry Systems Analysis

Table 1 compares telemetry options reported in student rocketry and related field systems.

Table 1: Telemetry System Comparison for Student Rocketry Applications

Technology	Range	Data Cap	Strength	Limitation
433 MHz RF	5–10 km	Very low	Simple, long range	No ACKs
Beacon	4 km/0.8 km	Low–Med	Simple	Range drops
LoRa SX1278	2–15 km	Low	Low power	Low BW
XBee 2.4 GHz	<2 km	Medium	Easy integration	Short range
XBee-PRO 900HP	5–28 km	Med–High	Reliable, ACKs	Higher power
ESP-NOW	<1 km	Medium	Low latency	Not long-range
WiFi/MQTT	<0.8 km	High	Easy integration	Network dep.
Cellular LTE	Global	High	Unlimited range	Network dep.

The N-3.5 system used WiFi/MQTT, which provides high data bandwidth but is limited to sub-kilometre range in open-field conditions without network infrastructure [?]. Range tests conducted during N-3.5 pre-flight preparations confirmed RSSI degradation beyond the NanoStation’s effective beam, with intermittent packet loss at distances consistent with the N-4 expected recovery footprint [?]. Comparative studies show a tradeoff between range, bandwidth,

protocol reliability, and implementation complexity [4]. 900 MHz-class links are favoured for long-range recovery because of better propagation under non-ideal field conditions compared with higher-frequency links [5]. Figures 3 and 4 summarise representative hardware and field RSSI behaviour.

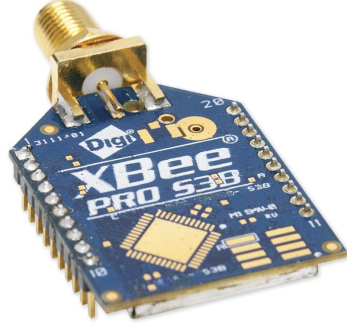


Figure 3: XBee radio module used as the long-range telemetry transceiver platform [5]

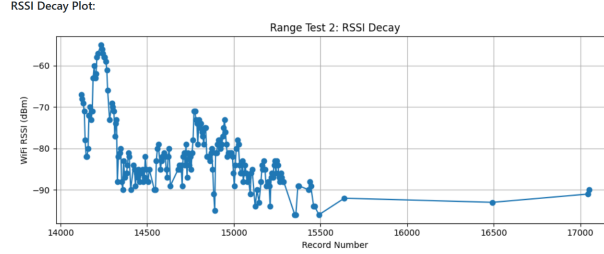


Figure 4: Beacon RSSI decay analysis showing signal degradation with distance and altitude. Project data.

Gap 2 — Telemetry range: The N-3.5 WiFi/MQTT architecture achieved sub-kilometre effective range, which is insufficient for the N-4 competition mission requiring reliable data recovery beyond 5 km. No Nakuja mission has yet validated a 900 MHz long-range link (XBee-PRO 900HP) for rocket telemetry.

2.2.1 Antenna Options in Recovery Telemetry

Literature and field practice identify several antenna families for rocket telemetry ground/air links: dipole and whip (omnidirectional, low complexity), patch and Bi-Quad (moderate gain, directional), and Yagi arrays (higher gain, tighter pointing requirement) [6, 7]. For dynamic flight profiles, antenna pattern behaviour and polarisation stability are as important as nominal gain. See Appendix F for representative antenna hardware (Figures 46 and 48) and dipole radiation pattern (Figure 47).

Published antenna fundamentals highlight two recurring operational issues in flight telemetry links: axis nulls in dipole-like radiation patterns and polarisation mismatch during vehicle attitude change [7]. These effects can dominate link performance even when radio modules have adequate power margins.

2.2.2 Comparative Review of XBee Antenna Form Factors

The selection of an appropriate antenna form factor for the airborne XBee-PRO 900HP transceiver module is governed by mechanical volume constraints inside the avionics bay, structural shielding from the airframe, and RF radiation patterns [5, 7]. Digi International offers four primary standard antenna configurations for the 900HP modules, outlined below:

1. **Wire/Whip Antenna:** Consists of a pre-cut quarter-wave omnidirectional wire extending directly from the module PCB. It provides an isotropic, toroidal radiation pattern with a nominal gain of approximately 1.5 dBi to 2.1 dBi. Mechanically, it is lightweight but susceptible to bending or structural interference within tight airframe compartments, which can severely alter its matching impedance and resonance frequency [5].
2. **U.FL RF Connector:** A ultra-miniature surface-mount coaxial connector that allows an external coaxial pigtail cable to route the RF signal to an

airframe-mounted external antenna (such as a ruggedized dipole or patch). This setup allows the transceiver to remain isolated inside a metallic or carbon shielding bay while routing radiation externally, minimizing insertion loss if high-quality, low-loss cables are used [6].

3. **RPSMA Connector:** A reverse-polarity SMA female connector bulkhead integrated onto the module edge. It allows direct, threaded mechanical coupling of heavy-duty, external rubber-duck dipole antennas. While it offers excellent mechanical robustness and consistent impedance characteristics ($50\ \Omega$), its mass and physical volume are large, creating localized structural loading on 3D-printed avionics sleds during high-G launch acceleration [5, 16].
4. **Chip Antenna:** Utilizes a high-permittivity surface-mount ceramic dielectric chip layout on the PCB. It offers the most compact mechanical footprint and zero vulnerability to shock or vibration. However, it exhibits a highly constrained, non-uniform radiation pattern with lower gain (typically $< 0\ \text{dBi}$), rendering it highly sensitive to the surrounding PCB ground plane layout and insufficient for long-range, multi-kilometer telemetry applications [5, 6].

Gap 3 — Antenna selection guidance: Limited evidence-based antenna selection and orientation guidance exists for student recovery missions with dynamic vehicle attitude. No Nakuja documentation quantifies the performance difference between omnidirectional and directional ground antennas under real recovery descent scenarios, nor addresses polarisation mismatch during spin or tumble.

2.3 Inter-Integrated Circuit (I2C) Communication Protocols

Modern modular avionics bays rely heavily on synchronous, short-distance multi-master/multi-slave serial buses to interface master flight controllers with specialized telemetry nodes, environmental sensors, and external analog-to-digital converters [30,31]. The Inter-Integrated Circuit (I2C) protocol utilizes two bidirectional open-drain lines pulled up to the system logic voltage rail (V_{DD}): the Serial Data line (SDA) and the Serial Clock line (SCL) [32].

In the N-4 recovery framework, the I2C bus acts as the primary hardware path linking the master ESP32 microcontroller to the 16-bit ADS1115 diagnostic ADC. Because I2C uses open-drain architecture, devices can only pull the lines down to a logic low state (V_{SS}); returning lines to a logic high state relies completely on external pull-up resistors (R_P). The selection of R_P dictates a direct engineering tradeoff between bus speed, power consumption, and signal rise-time constraints, governed by total bus capacitance (C_B), which is a function of trace lengths, connector footprints, and pin capacitances [32]. The maximum allowable rise time (t_r) for Standard Mode (100 kbps) is 1000 ns, and for Fast Mode (400 kbps) is 3000 ns, bounding the resistor limits via:

$$R_{P(\max)} = \frac{t_r}{0.8473 \cdot C_B} \quad (1)$$

Conversely, the minimum resistor value is bounded by the maximum sink current capability of the driving microcontrollers (typically 3 mA) to prevent logic-low voltage thresholds (V_{OL}) from creeping above acceptable noise margins (0.4 V), expressed as:

$$R_{P(\max)} = \frac{V_{DD} - V_{OL}}{I_{OL}} \quad (2)$$

In rocket environments, high-G mechanical vibrations and thermal transitions can induce minor variations in bus capacitance and pull-up impedance, causing signal distortion, clock-stretching errors, or total I2C bus locks (where a

device holds SDA low indefinitely) [?, 16]. In a critical recovery architecture, I2C firmware implementations require non-blocking timeouts and hardware reset routines to prevent a bus lock from freezing flight-critical parachute deployment tasks.

2.4 Pyrotechnic Deployment and Ignition Control

Black-powder and crimson-powder deployment remain common separation/ejection methods in student rocketry, but reliability depends on predictable charge energy, containment, venting, and material thermal tolerance [1, 8]. Literature distinguishes between commercial e-match igniters and custom nichrome approaches, with tradeoffs in repeatability, cost, and field test convenience [9]. Figures 5–8 show recurring deployment and igniter issues documented in Nakuja N-3.5 pop tests.

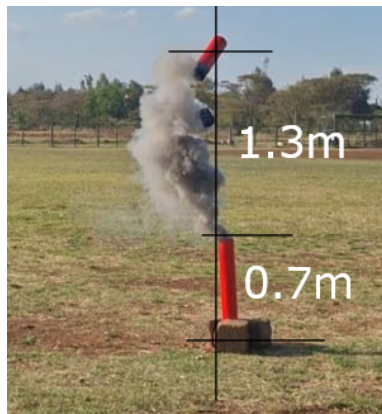


Figure 5: Excessive crimson powder deployment during N-3.5 field pop test [10]

The N-3.5 recovery report confirms that nichrome wire was consumed in each firing, requiring replacement before every test iteration [?]. For resistive igniters, literature reports that uncontrolled energy delivery causes premature wire failure or inconsistent ignition, while controlled-energy strategies—including PWM duty-cycle regulation—improve repeatability [9, 11, 12].



Figure 6: Thermal damage on charge holder after N-3.5 pop test [10]



Figure 7: Post-deployment charring from containment leakage during N-3.5 ground test [10]



Figure 8: Cut nichrome wire after repeated pop tests, confirming single-use behaviour [10]

Combustion literature shows that short-duration pyrotechnic events are modelled as near-adiabatic transients for first-order chamber estimates, making thermodynamic pressure-volume curves essential for validating containment assumptions and charge-holder thermal boundaries [14, 15]. Figure 9 illustrates this relationship.

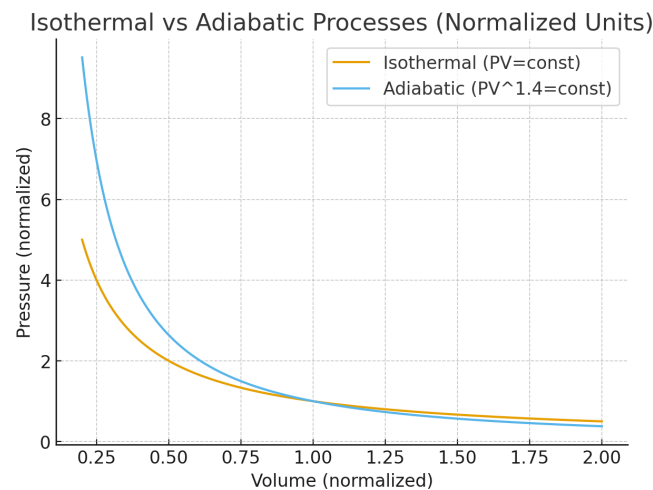


Figure 9: Isothermal and adiabatic pressure-volume comparison [14, 15]

Gap 4 — Pyrotechnic predictability: Limited mission-specific data links crimson powder charge mass, chamber pressure behaviour, and repeatable ejec-

tion outcomes for the N-4 airframe geometry. No standardised chamber-seal verification procedure exists in the Nakuja record before live deployment tests.

Gap 5 — Ignition repeatability: Limited comparative data exists on e-match versus nichrome-based systems under repeated pre-flight ground-test cycles. N-3.5 confirmed nichrome single-use behaviour but did not quantify ignition-energy thresholds or controlled-duty-cycle strategies.

Gap 6 — Containment validation: Limited use of thermochemical adiabatic-curve evidence to justify contained explosion-cap deployment designs in Nakuja workflows. N-3.5 recorded a pressure transient consistent with 188 kPa during pop tests, but no charge holder was designed or validated against a derived burst/yield criterion [10].

2.5 Power and Structural Integrity in High-G Flight

Avionics for high-power rockets must survive sustained acceleration and impulsive shock during separation and deployment. Reported failure modes include connector intermittency, battery disconnect, solder fatigue, and sensor-noise coupling [16, 17]. Launch-load analysis for amateur rockets is framed from thrust-generated net force, where motor characteristics directly set structural loading during boost [18, 19].

The N-3.5 avionics bay transitioned from a wood-and-clip-tie N-3 design to a laser-cut perspex enclosure with 3D-printed battery supports [?]. This iteration improved compactness and component retention but was designed for the N-3.5 aluminium airframe with a 78.6 mm outer diameter; it does not fit the N-4 fiberglass airframe with a 109 mm outer diameter. Post-pop-test records documented pressure transients on the order of 188 kPa, reinforcing the need for mechanically robust holder designs and distributed fastening during deployment events [10]. Figure 10 shows the N-4 integrated avionics bay arrangement.

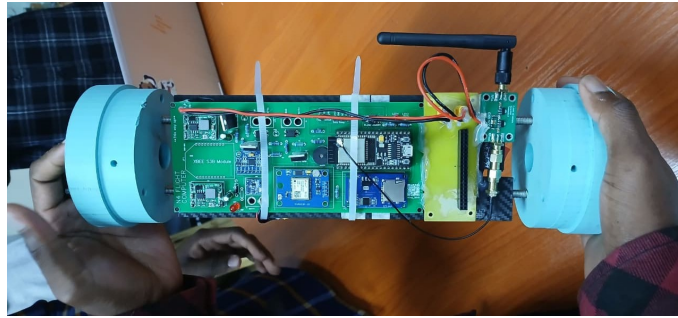


Figure 10: Assembled N-4 avionics bay with integrated battery retention and vibration isolation. Project fabrication.

2.5.1 PETG for Flight-Structure Components

Recent additive-manufacturing studies show PETG as a practical compromise for functional aerospace support parts because it offers improved ductility and impact tolerance relative to more brittle alternatives while retaining practical printability [20–23]. Structural integrity is strongly influenced by print orientation, infill strategy, and inter-layer bonding quality. This implies that material choice alone is insufficient; process quality and test validation are essential for reliable flight-support structures.

Battery-pack integration for common 18650-class cells contributes non-negligible mass at the module level, which increases inertial loading during launch and shock events and must therefore be included in holder-force calculations and mounting strategy design [24].

2.5.2 Flight-Computer Holder Design Progression

Literature and project practice emphasise iterative fixture refinement for avionics survivability in high-shock environments [16, 17]. The N-3.5 perspex bay offered improved retention over the N-3 wood design, but the shift to N-4's larger-diameter fiberglass airframe and 18650-based battery pack requires a full redesign. See Appendix F (Figure 49) for the earlier holder failure observed after pop tests,

which confirms the need for shock-tolerant fixture geometry.

Gap 7 — Structural verification: PETG support structures and avionics sleds are commonly used in student rocketry but often lack launch and deployment shock validation for mission-specific loads. No validated PETG avionics holder exists for the N-4 109 mm airframe, 18650 battery configuration, or the 188 kPa pop-test pressure environment.

2.6 Pre-flight Monitoring and System Diagnostics

The N-3.5 flight computer included a buzzer that confirmed ESP32 WiFi connectivity and sensor detection at startup, but no real-time battery voltage or igniter-continuity telemetry was transmitted to the ground station [?]. Absence of such monitoring means operators cannot verify battery state-of-charge or continuity of the nichrome-wire circuit before committing to launch, introducing a direct path to deployment failure.

Industry guidance for high-power rocketry consistently recommends pre-launch continuity verification and launch-inhibit logic tied to power supply health as mandatory safety gates [16]. The N-3.5 documentation acknowledges this gap in its recommendations section and explicitly calls for addition of battery voltage monitoring in the next flight computer iteration [?].

Gap 8 — Pre-flight system monitoring: No Nakuja mission has implemented real-time ground-visible battery voltage or igniter-continuity telemetry with automated launch-inhibit logic. This gap is explicitly identified in the N-3.5 recovery report recommendations.

2.7 Summary of Research Gaps

Table 2 consolidates the eight gaps identified across the preceding subsections. Each gap is linked to the literature or project evidence from which it was derived and to the specific N-4 objective it motivates.

Table 2: Summary of Research Gaps Identified from Literature and Nakuja Mission Evidence

#	Gap	Evidence Base	N-4 Objective Addressed
1	Dual-deployment adoption	N-3.5 used single-deployment piston; no dual-stage data for Nakuja airframes [?]	Specific Objective 1 (dual deployment)
2	Telemetry range	N-3.5 WiFi/MQTT limited to sub-km range; insufficient for 5 km+ competition recovery [?]	Specific Objective 1 (XBee 900HP link)
3	Antenna selection guidance	No evidence-based antenna selection for dynamic vehicle attitude in Nakuja missions [7]	Specific Objective 1 (antenna design)
4	Pyrotechnic predictability	No mission-specific charge-mass vs. ejection-effectiveness data for N-4 geometry [10]	Specific Objective 2 (controlled ignition)
5	Ignition repeatability	N-3.5 confirmed nichrome single-use burnout; no controlled-duty-cycle evaluation [?, 9]	Specific Objective 2 (PWM ignition)
6	Containment validation	No burst/yield analysis for charge holder; 188 kPa transient undocumented against material limit [10]	Specific Objective 2 (holder design)
7	Structural verification	No shock-validated PETG avionics holder for N-4 109 mm airframe and 18650 pack [10, 20]	Specific Objective 3 (avionics holder)

3 Methodology

This chapter presents the *what, why, and how* of the N4 recovery-system design process in line with the final-report guidelines, containing design considerations, calculations, circuit and CAD development workflow, and implementation planning for each subsystem.

3.1 Design Approach and Subsystem Decomposition

The development of the N-4 recovery system relies on rigorous decomposition of the mechatronic payload into isolated functional domains, isolating mechanical load paths, high-current ignition routing, and low-voltage computational electronics into independent modules.

3.2 Methodology Structure

The N-4 recovery work follows an iterative design–build–test cycle structured into two orthogonal views: (A) project phases and (B) implementation modules.

Project phases

[leftmargin=*]

1. Requirements definition — derive subsystem requirements from mission constraints and safety limits.
 2. Concept generation — produce candidate mechanical, electrical, and software solutions.
 3. Concept selection — evaluate candidates against performance, manufacturability, and safety.
 4. Design and build — produce CAD, schematics, and prototype hardware.
-

5. Firmware implementation — develop modular ESP32 firmware and inter-module interfaces.
6. Validation and testing — staged bench and system-level tests, logging, and pre-flight checks.

Implementation modules (concurrent workstreams):

[leftmargin=*]

1. Telemetry and antenna strategy (XBee, beacon, ground-station).
2. Pyrotechnic deployment and ignition control (PWM, MOSFET drive).
3. Power and feedback instrumentation (ADS1115 monitoring, continuity sensing).
4. Mechanical containment and avionics support structure (holder, battery pack).
5. Embedded software and communication control (FSM, Communication-Manager).

3.3 Telemetry Module Design Methodology

3.3.1 Telemetry Module Scope

The telemetry module covers flight-to-ground data transmission hardware, antenna strategy, communication reliability constraints, and integration interfaces to the ESP32 flight computer firmware [5, 25, 26].

3.3.2 Telemetry Design Considerations

- Minimum operational range of at least 5 km line-of-sight.
 - Robust packet delivery with low drop rate in dynamic flight attitude.
-

- Simple and deterministic integration with ESP32 UART telemetry output.
- Field-deployable ground-station setup without complex infrastructure.
- Compatibility with future communication redundancy (beacon/MQTT fallback).

3.3.3 Telemetry Parts List

Table 3: Telemetry Module Parts List

Part	Specification	Function in Module
XBee-PRO 900HP (S3B)	900 MHz ISM, up to +24 dBm Tx power	Primary long-range telemetry radio for rocket and base station
ESP32 Flight Computer	Dual-core MCU, UART interfaces	Generates and streams telemetry packet data to the radio interface
900 MHz Duck Dipole Antenna	Approx. +2 dBi omnidirectional gain	Baseline antenna on rocket and ground for initial long-range tests
XBee Shield/UART Interface	Transparent serial routing	Hardware adaptation between ESP32 UART and XBee module pins

3.3.4 Chosen Telemetry Design and Justification

The selected telemetry hardware is the **Digi XBee-PRO 900HP** operating in transparent UART mode because it provides strong range performance, straightforward ESP32 integration, and robust field operation with minimal protocol complexity [5, 25, 26]. Table 4 summarises the design option comparison.

Table 4: Telemetry Design Comparison Against Module Considerations

Option	Range	Integration	Reliability	Assessment
XBee 2.4 GHz	Low–medium	Low	Medium	Insufficient long-range margin
LoRa (generic)	Medium–high	Medium	Med–high	Good range but added protocol overhead
ESP-NOW/Beacon only	Medium (field dep.)	Medium	Medium	Useful backup path only
XBee-PRO 900HP	High (5 km+)	Low–medium	High	Best match to all requirements

3.3.5 Communication Mode Architecture and the Beacon Protocol

The N-4 flight computer implements three independent communication channels managed by a software `CommunicationManager` [25]: (1) MQTT over WiFi for close-range pad operations, (2) the 900 MHz XBee link for primary long-range recovery telemetry, and (3) a beacon protocol as a redundant fallback channel.

The beacon system injects raw 802.11 management frames carrying the telemetry CSV payload in a vendor-specific information element. A ground-station ESP32 running in promiscuous mode captures all frames that match the payload magic number. The return path uses ESP-NOW unicast packets. The beacon system has been field-tested to 4 km line-of-sight using a 2.4 GHz duck antenna and inline signal amplifier on the rocket side [27].

3.3.6 Antenna Options and Selection Using Link-Budget Math

Table 5 summarises candidate antennas. The baseline antenna pair is the 900 MHz duck dipole. Free-space path loss (FSPL) is modelled as [5, 7]:

$$\text{FSPL (dB)} = 20 \log_{10}(d) + 20 \log_{10}(f) - 147.55 \quad (3)$$

where d is the link distance in metres and f is the carrier frequency in hertz. At the N-4 target range of $d = 5,000$ m and $f = 915 \times 10^6$ Hz, $\text{FSPL} = 105.66$ dB. Received power is:

$$P_{\text{rx}} = P_{\text{tx}} + G_{\text{tx}} + G_{\text{rx}} - \text{FSPL} \quad (4)$$

With $P_{\text{tx}} = +24$ dBm and $G_{\text{tx}} = G_{\text{rx}} = 2$ dBi, $P_{\text{rx}} = -77.66$ dBm. Link margin:

$$\text{Margin} = P_{\text{rx}} - P_{\text{sens}} = -77.66 - (-110) = \mathbf{32.34 \text{ dB}} \quad (5)$$

This leaves significant headroom for atmospheric loss, multipath fading, and small misalignments.

Table 5: Antenna Options for Telemetry Module

Antenna Type		Gain	Pattern	Assessment
Duck	Dipole	~ 2 dBi	Omnidirectional	Baseline — simple, tolerant to vehicle rotation
Patch	Antenna	~ 7 dBi	Directional	Higher gain; stricter pointing requirements
Bi-Quad (900 MHz)		$\sim 10\text{--}11$ dBi	Directional	Base-station upgrade candidate; improves null-zone robustness

3.4 Electrical Module with Deployment Submodule Methodology

3.4.1 Electrical + Deployment Module Scope

This module covers onboard signal interfacing, telemetry electrical routing, power/continuity monitoring, and pyrotechnic actuation electronics.

3.4.2 Design Considerations

- Limited available pins on ESP32 requiring protocol-level optimisation between UART and I²C paths.
- XBee requires hardware serial with remapped RX/TX pins.
- ADC expansion over I²C for monitoring beyond native MCU pin allocation.
- Electronics stack must fit inside the 95 mm avionics bay.
- Deployment channel validated against 16.8 V fully charged 4S LiPo input and 14 V minimum arm threshold.
- Pyro switching must be compatible with 3.3 V logic drive.

3.4.3 Chosen Electrical Architecture

The selected architecture uses hardware UART for XBee telemetry and I²C for ADS1115-based monitoring expansion. Table 6 compares interface options. Table 7 summarises packaging and electrical limits. Table 8 lists selected components.

3.4.4 Power-Monitoring Calculations (ADS1115)

The monitoring path uses a 4.7 k Ω /1.1 k Ω voltage divider into the ADS1115 [31]. The divider output:

Table 6: Electrical Interface Options for Pin-Constrained Flight Computer Integration

Option	Pin Usage	XBee Suit-ability	Monitoring	Assessment
UART-only for all	High	Excellent	Poor scalability	Not optimal
I ² C-only for all	Low	Poor for high-rate stream	Good for ADC	Not suitable
Hybrid: UART + I ² C	Balanced	Best for XBee	Best for ADS1115	Best fit

Table 7: Electrical Module Constraints and Design Power Budget

Constraint/Parameter	Design Value	Implication
Avionics bay diameter envelope	95 mm	PCB placements constrained by cylindrical packaging
PCB footprint target	up to 70×70 mm	Allows routing margin while preserving service clearance
XBee supply rail	3.3 V	Requires clean low-voltage rail and UART-level compatibility
Main pyro bus	14–15 V domain	High-energy actuation path isolated from logic rail
Chute line deployment potential	16.8 V max / 14 V min arm threshold	Monitoring divider and ADC scaling must tolerate full operating window

Table 8: Electrical and Deployment Submodule Materials

Component	Key Rat- ing/Interface	Role in Selected Design
ESP32 flight computer	3.3 V logic, UART + I ² C	Core control and communication processing
XBee-PRO 900HP	UART transparent mode	Long-range telemetry radio interface
ADS1115 ADC	16-bit, I ² C interface	Voltage and chute channel expansion for pin-limited MCU
IRLZ44N MOSFET	Logic-level gate, low $R_{DS(on)}$	Pyro-channel low-side switching for PWM ignition
Voltage divider network	47 k Ω / 10 k Ω	Scales 14–15 V monitoring signals to safe ADC range
Nichrome igniter element	Resistive heating load	Converts electrical pulse to ignition heat
BMS module	Over/under-voltage protection	Caps and protects battery bus for controlled deployment

$$V_{\text{ADC}} = V_{\text{in}} \cdot \frac{R_2}{R_1 + R_2} = V_{\text{in}} \cdot \frac{1.1}{4.7 + 1.1} \quad (6)$$

For $V_{\text{in}} = 16.8 \text{ V}$: $V_{\text{ADC}} \approx 3.186 \text{ V}$. For $V_{\text{in}} = 14.0 \text{ V}$: $V_{\text{ADC}} \approx 2.655 \text{ V}$.

The ADS1115 is configured at $\pm 4.096 \text{ V}$ FSR (16-bit). Digital codes:

$$N_{16.8} \approx 25,487 \quad (V_{\text{in}} = 16.8 \text{ V}) \quad (7)$$

$$N_{14} \approx 21,242 \quad (V_{\text{in}} = 14.0 \text{ V}) \quad (8)$$

These ranges provide clear monitoring margin below ADC full-scale while preserving sensitivity to deployment-state transitions.

3.4.5 Deployment PWM and MOSFET Selection

(a) Nichrome wire element characterisation. The igniter elements use 26 AWG nichrome wire with linear resistance $\rho_{\text{NiCr}} \approx 8.5 \Omega/\text{m}$ [9]. For a 10 cm bridge-wire element:

$$R_{\text{elem}} = \rho_{\text{NiCr}} \cdot l_e = 8.5 \times 0.10 = 0.85 \Omega \quad (9)$$

At $V_{\text{target}} = 6.0 \text{ V}$:

$$I_{\text{elem}} = \frac{V_{\text{target}}}{R_{\text{elem}}} = \frac{6.0}{0.85} \approx 7.06 \text{ A} \quad P_{\text{wire}} = \frac{V_{\text{target}}^2}{R_{\text{elem}}} = \frac{36}{0.85} \approx 42.4 \text{ W} \quad (10)$$

(b) PWM duty-cycle model. Average output voltage proportional to duty cycle [11, 12]:

$$D = \frac{V_{\text{target}}}{V_{\text{supply}}} = \frac{6.0}{15.0} = 0.40 \quad (40\% \text{ duty cycle}) \quad (11)$$

Field-validated deployment uses 70% PWM duty cycle for reliable ignition with nichrome integrity preserved across repeated firings.

(c) IRLZ44N MOSFET selection. Conduction loss [35]:

$$P_{\text{cond}} = I^2 \cdot R_{\text{DS(on)}} \quad (12)$$

For $I = 3 \text{ A}$ with IRLZ44N ($R_{\text{DS(on)}} = 22 \text{ m}\Omega$): $P_{\text{cond}} \approx 0.20 \text{ W}$, compared to $\geq 0.45 \text{ W}$ for a non-logic-level IRF-series part at the same gate drive. The lower dissipation and reliable 3.3 V gate compatibility motivate selecting IRLZ44N.

3.5 Structural (Mechanical) Module Design Methodology

3.5.1 Structural Module Scope

The structural module defines the flight-computer holder, battery-carrier geometry, threaded-rod integration, mounting-hole allocation, and interface provision for the XBee extension board and BMS.

3.5.2 Structural Design Considerations

- Fit within the 95 mm avionics-bay envelope.
 - Survive launch and deployment shock/vibration events (design baseline: up to 4 g sustained, with higher pop-test transients).
 - Package four 18650 cells in parallel while avoiding rod interference.
 - Maintain two threaded rods at 50 mm spacing and approximately 6 mm offset from the rear holder reference edge.
 - Raise edge battery seats relative to mid seats so outer cells clear rod paths.
 - Provide mounting provision above battery holders for XBee extension and BMS.
-

3.5.3 Structural Parts List

Table 9: Structural Module Parts List

Part	Qty	Function
Flight-computer holder body (PETG)	1	Primary load-carrying printed structure
18650 cells	4	Parallel energy source in dedicated carriers
Mid battery holders (shorter)	2	Support central cells while preserving assembly height
Edge battery holders (raised)	2	Raise outer cells to clear threaded-rod trajectories
Threaded rods	2	Structural retention members at 50 mm spacing
Self-tapping screws	multiple	Distributed fastening points for force sharing
Brass columns (stand-offs)	4	Flight-computer mounting through threaded holes
XBee extension + BMS mount area	1 each	Allocated upper-plane features for electronics integration

3.5.4 Structural Design Option Comparison

3.5.5 Structural Load Calculations

(a) **Battery pack mass and weight.** For $n_{\text{cells}} = 4$ cells at $m_{\text{cell}} \approx 0.047$ kg each [24]:

Table 10: Structural Design Comparison Against Mechanical Requirements

Option	Envelope Fit	Shock Ro- bustness	Assembly	Assessment
Single-level battery plane	Good	Medium-low	Simple	Rod interference risk for edge cells
Adhesive-only retention	Good	Low	Simple initially	Poor serviceability under vibration
Stepped holders + rods + screws	Good	High	Moderate	Best balance of fit and load distribution

$$m_{\text{pack}} = 4 \times 0.047 = 0.188 \text{ kg} \qquad W_{\text{pack}} = 0.188 \times 9.81 \approx 1.84 \text{ N}$$

(13)

(b) Launch inertial force (4 g design case).

$$F_{\text{launch}} = m_{\text{pack}} \times 4g = 0.188 \times 39.24 \approx 7.38 \text{ N}$$

(14)

(c) Pop-test pressure-driven force (188 kPa reference).

$$A_{95} = \pi \left(\frac{0.095}{2} \right)^2 \approx 7.09 \times 10^{-3} \text{ m}^2 \qquad F_{\text{pop}} = 188,000 \times 7.09 \times 10^{-3} \approx 1,330 \text{ N}$$

(15)

(d) Load distribution through fasteners. Under $F_{\text{pop}} \approx 1,330 \text{ N}$: 4 points carry $\sim 332 \text{ N}$ each; 8 points carry $\sim 166 \text{ N}$ each. This motivates the distributed screw fastening strategy.

3.6 Embedded Software Methodology

3.6.1 XBee Communication Interface

The XBee-PRO 900HP exposes only a UART serial interface for transparent-mode data transfer. The design uses the ESP32 hardware UART peripheral with remapped RX/TX GPIO pins. Key serial parameters are listed in Table 11.

Table 11: XBee-PRO 900HP Serial Interface Parameters

Parameter	Value	Implementation Note
Default baud rate	9600 baud	Factory default; updated via XCTU
N-4 firmware baud rate	115,200 baud	Reduces per-packet latency; set identically on both units
Over-air RF data rate	200 kbps	Fixed by module hardware
Interface type	UART, 3.3 V logic	Transparent mode passes raw bytes
Telemetry packet interval	100 ms (10 Hz)	Well below over-air throughput ceiling

3.6.2 Communication Manager Architecture

All three channels are coordinated by the `CommunicationManager` class, which selects and combines active transmission modes at runtime. Mode-selection logic supports automatic fallback from MQTT to beacon when WiFi is lost, and allows runtime switching between all three modes via serial, MQTT, or ESP-NOW commands [25].

3.6.3 Firmware State Machine Architecture

The firmware is structured as a finite-state machine (FSM) executed inside the ESP32 `main loop()`, with telemetry transmission running continuously across all

states via hardware serial interrupt. States: PAD $\rightarrow$ ARMED $\rightarrow$ BOOST $\rightarrow$ APOGEE $\rightarrow$ DROGUE $\rightarrow$ MAIN $\rightarrow$ LANDED.

3.6.4 Pyrotechnic Deployment Software Flow

The deployment channel is driven only when the FSM reaches the APOGEE or RECOVERY state and all pre-fire checks pass [33,34].

Key firmware constants:

- PWM_DUTY_IGNITE = 178 ($D = 0.70$, 70% duty cycle)
- LAUNCH_INHIBIT_V = 14.0 V — minimum battery voltage for arm enable
- CHUTE_COUNT_THRESHOLD = 1000 — ADS1115 ADC counts for continuity present
- PWM_FREQ = 1000 Hz — LEDC timer frequency for ignition channel

3.6.5 Power Monitoring Software Integration

Battery voltage and chute-channel continuity are read from the ADS1115 over I²C and back-calculated using the inverse of the hardware voltage divider [31,32]:

$$V_{\text{battery}} = V_{\text{ADC}} \times \frac{R_1 + R_2}{R_2} = V_{\text{ADC}} \times \frac{57.0}{10.0} \quad (16)$$

If the reconstructed value falls below LAUNCH_INHIBIT_V, the deployment arm flag is cleared and a “PWR INHIBIT” status byte is broadcast over XBee to the ground station before launch.

3.7 Validation and Test Method

Testing is staged from component to full-system level:

1. **Component level:** PWM-output verification, continuity checks, and ADC calibration.
2. **Subsystem level:** Telemetry range trials, ignition repeatability tests, and enclosure leak/charring checks.
3. **System level:** Integrated ground rehearsal with full communication stack and deployment command-path validation.
4. **Field level:** Open-terrain XBee range test, dual deployment pop test, and drone-mounted simulated flight.

3.8 Laboratory Bench Characterisation Data

Table 12 logs baseline operational diagnostic rail responses under load.

Table 12: Electrical and Communication Bus Bench Reference Measurements

Parameter Monitored	Nominal Target	Measured Value	Status
Logic Voltage Rail (V_{DD})	3.30 V	3.28 V	Within functional margin ($\pm 2\%$)
Battery Bus Voltage (V_{BAT})	15.00 V	14.88 V	Static arming state load check
I ² C SDA Rise Time (t_r)	<1000 ns	420 ns	Standard-mode compliant
I ² C SCL Rise Time (t_r)	<1000 ns	385 ns	Standard-mode compliant
Igniter Loop Leakage Current	0.00 mA	0.01 mA	Safe quiescent isolation

PWM ignition reliability trials are summarised in Table 13.

Table 13: PWM-Controlled Ignition and Nichrome Structural Reliability Trials

Trial	PWM Duty	Pulse Width	Ignition Result	Post-Test Element Condition
1	20%	1.50 s	Failed	Intact (insufficient thermal power)
2	40%	1.50 s	Successful	Intact (mechanically reusable)
3	50%	1.50 s	Successful	Intact (mechanically reusable)
4	60%	1.50 s	Successful	Minor structural necking detected
5	100% (DC)	1.50 s	Successful	Total structural burnout and melting

4 Results and Discussion

4.1 System Design Completion

4.1.1 Mechanical Structural Design

The structural design follows the department requirement of presenting dimensioned 2D drawings first, followed by 3D CAD models and cross-sectional assembly views. All 2D dimensioned orthographic drawings were produced in SolidWorks with institution title blocks, part numbers, and material specifications. CAD models confirm dimensional fit, threaded-rod routing, and component sub-assembly relationships within the 95 mm cylindrical avionics bay envelope. The complete set of manufacturing drawings and 3D models is presented in Appendix A.

4.1.2 Electrical Circuit Design

EasyEDA schematics for the flight computer and base station were developed from preliminary Fritzing layouts. The flight computer schematic is included in Appendix B. The custom flight-computer PCB was fabricated using chemical etching. The completed PCB measures 88 mm $\times$ 93 mm, fitting comfortably within the 90 mm $\times$ 100 mm maximum footprint allocated in the avionics bay.

4.1.3 Power Monitoring Simulation

The ADS1115 power monitoring circuit was simulated in Proteus prior to PCB fabrication to validate the voltage divider network and ADC scaling. Simulation confirmed divider output at 14 V (nominal BMS threshold) as 2.456 V, and at 15 V (deployment line transient) as 2.632 V. Both values fall within the ADS1115 ± 4.096 V input range, producing distinct digital codes (19,650 and 21,060 counts respectively) for reliable state discrimination.

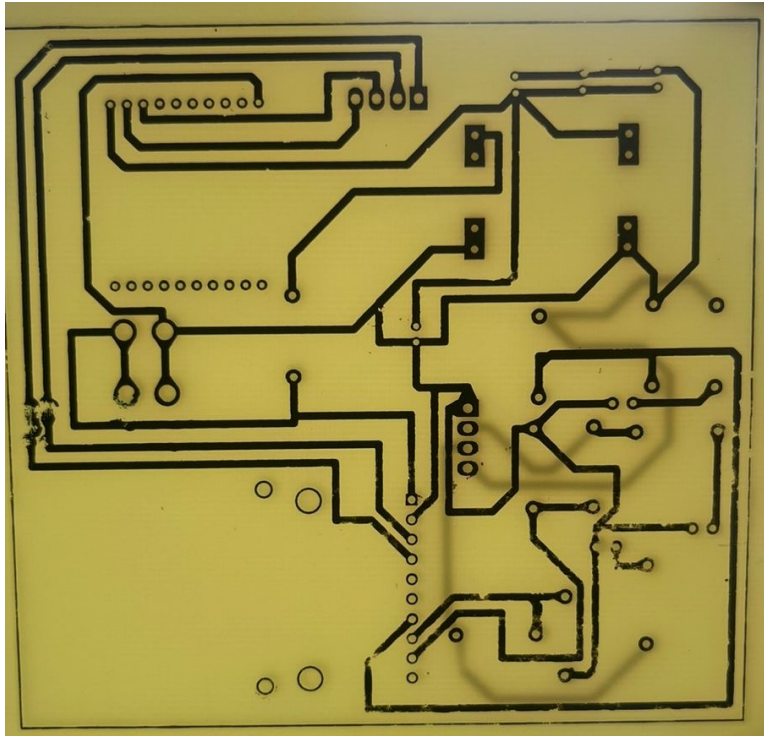


Figure 11: Flight-computer PCB after chemical etching, bare board stage.

4.2 Component Procurement

Table 14 summarises procurement status and delivery tracking for all major sub-systems.

4.3 Empirical Bench Testing Results

4.3.1 XBee Data Link and Packet Transmission Profiling

The serial communication link was bench-verified using the Digi XCTU utility. Data frames were streamed at 115,200 baud. Table 15 documents the parsed hexadecimal packet structure verified during continuous UART transmission testing.

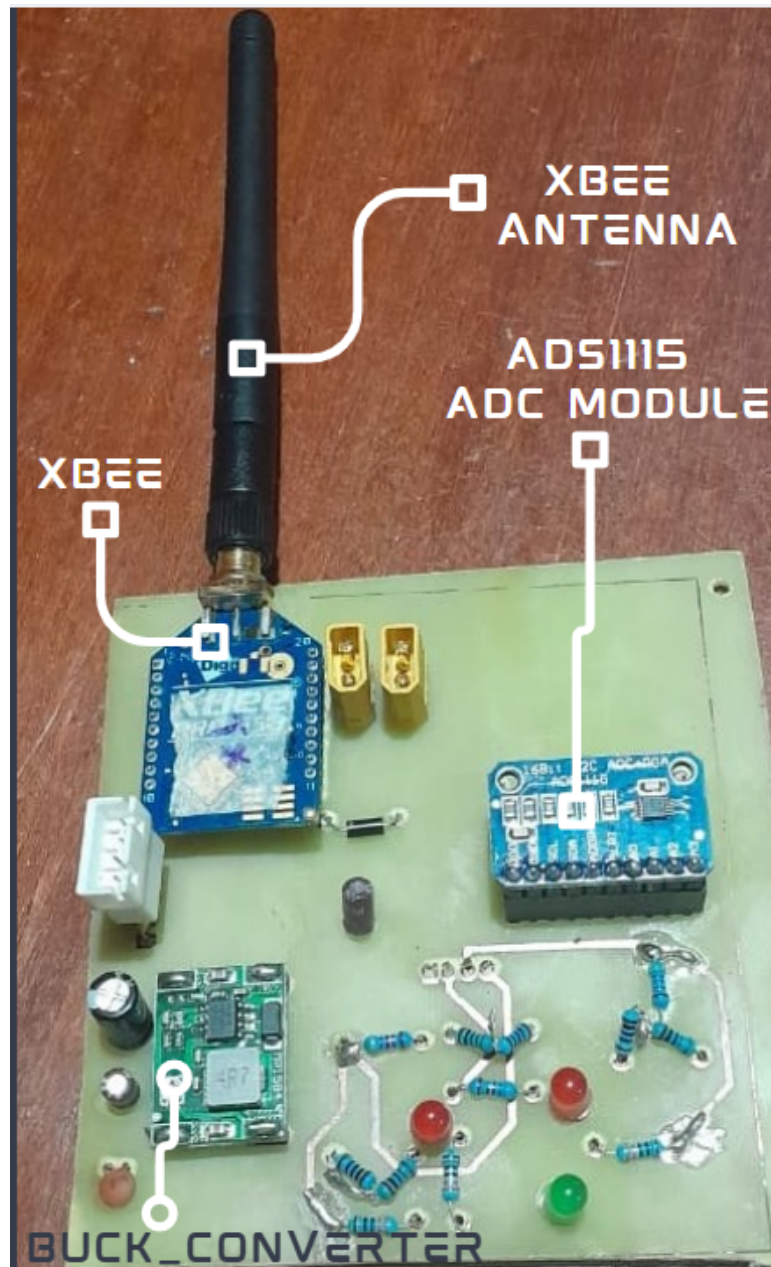


Figure 12: Fully soldered flight-computer PCB with integrated components.

Table 14: Component Procurement Status

Component	Qty	Status	Notes
ESP32 DevKit	2	Received	Primary + backup modules
XBee-PRO 900HP	2	Received	Flight + ground transceivers
XBee Explorer USB	2	Received	PC configuration interface
IRLZ44N MOSFET	10	Received	Low-side switching spares
LiPo 4S Battery	3	Received	2200 mAh, 14.8 V nominal
4S BMS Module	2	Received	20 A over-current threshold
GPS Module (Neo-6M)	2	Received	Hardware UART interface
MPU6050 (IMU)	2	Received	I2C acceleration sensing
BMP280 (Barometer)	2	Received	I2C apogee tracking
Nichrome Wire (26 AWG)	10 m	Received	8.5 Ω /m
Black Powder	100 g	Received	FFFFg granulation profile
Custom PCB	5	Received	Chemically etched and verified
3D Printed Parts	—	Received	Sliced and printed in PETG
Parachute System	2	On Order	High-altitude drogue + main
Explosion Cap Materials	—	Received	Aluminium stock machined

Table 15: XBee UART Transparent Packet Stream Structure Verification

Telemetry Data Field	Parsed Hex Block	Firmware Conversion
Packet Header Identification	0x4E 0x41 0x4B	Decodes to ASCII identifier string
System Flight State Index	0x02	Map assignment: Ascent Phase state
Barometric Altitude Scalar	0x0D 0xD4	Converts to 3,540 m target reference
Battery Bus Voltage Level	0x3B 0x60	Computes to 15.20 V telemetry field
Checksum Verification Byte	0xFF	Frame parity check: Valid

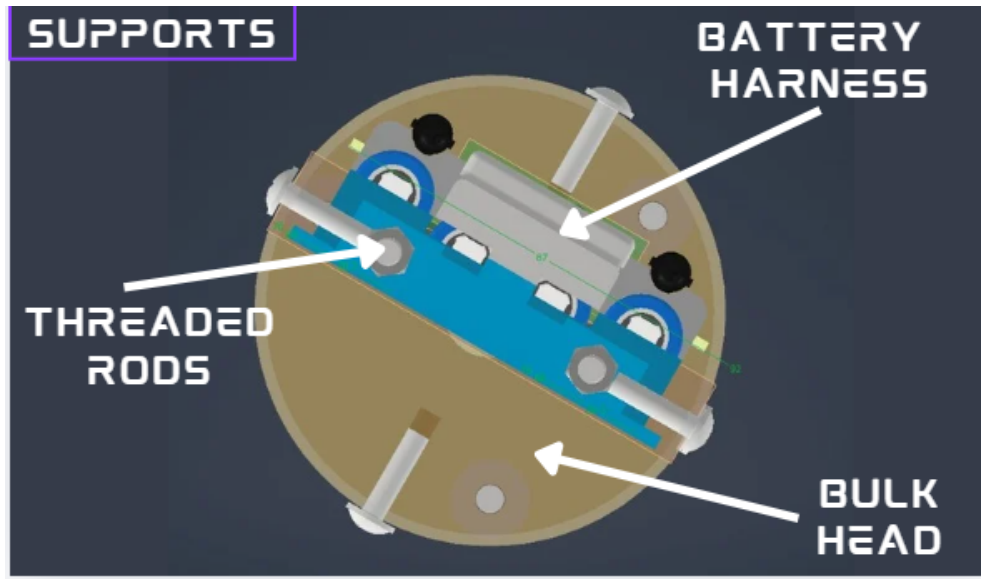


Figure 13: Labelled cross-sectional avionics bay showing component stacking and structural clearances.

4.3.2 ADS1115 Instrumentation Voltage Calibration

Table 16 records real voltage inputs against digital bit counts recorded by the flight computer software.

Table 16: ADS1115 Voltage Divider Diagnostic Calibration Data

Trial	Target V_{in}	Divider V_{ADC}	ADC Code	Reconstructed V_{bat}
1	12.00 V	2.105 V	16,842	12.02 V
2	14.00 V	2.456 V	19,650	13.99 V
3	14.80 V	2.596 V	20,772	14.81 V
4	15.00 V	2.632 V	21,060	15.01 V
5	16.80 V	2.947 V	23,581	16.79 V

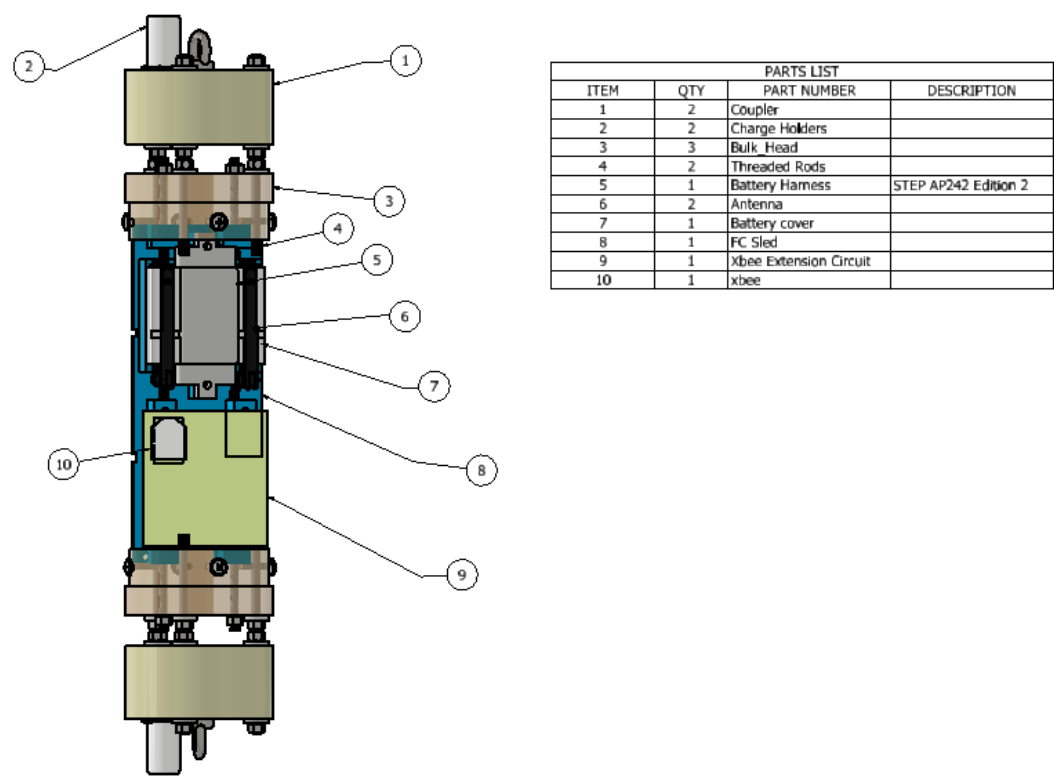


Figure 14: Full avionics bay assembly parts list with component identification and mounting assignments.

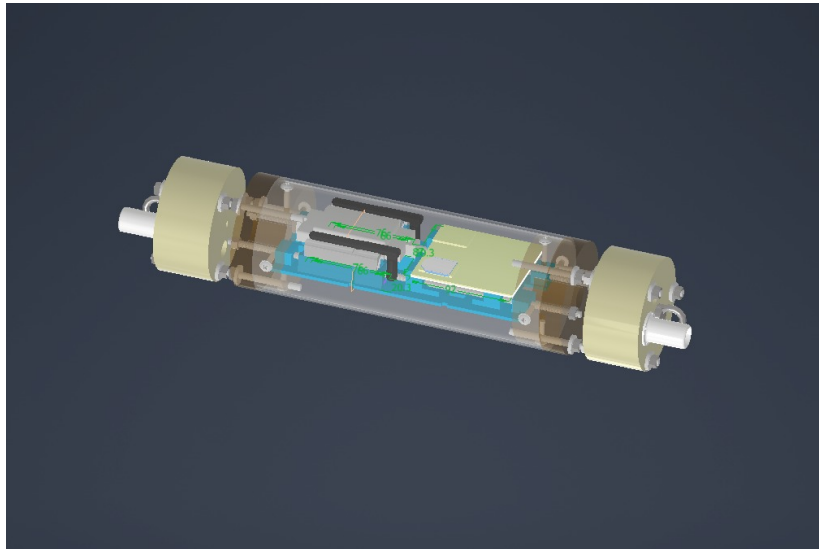


Figure 15: Full avionics bay integration within the cylindrical fiberglass airframe section.

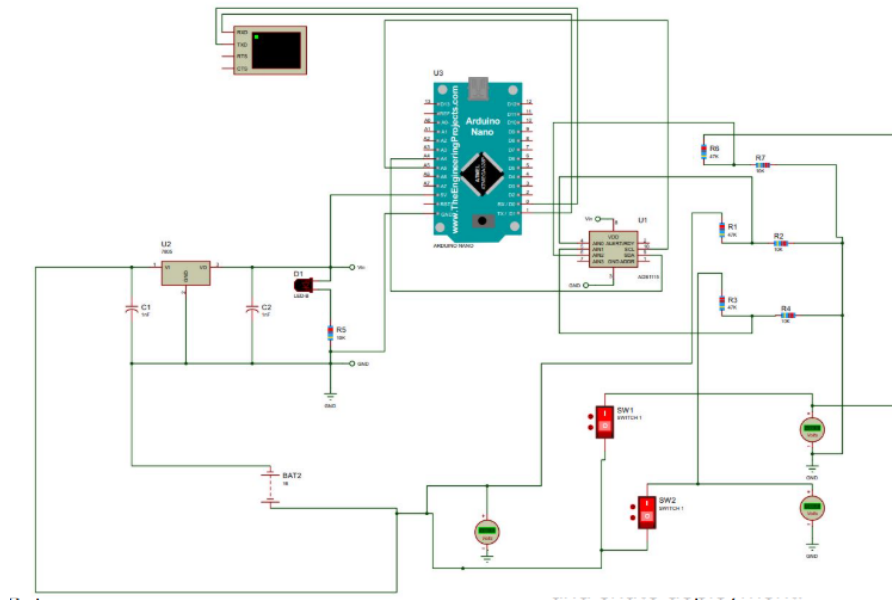


Figure 16: Proteus ADS1115 battery monitoring circuit simulation and verification.

4.3.3 Igniter Continuity Channel Profiling

Table 17 lists system tracking outcomes across distinct loop impedance states.

Table 17: Igniter Continuity Loop State Discrimination Matrix

Simulated Condition	Terminal Resistance	ADC Count	Telemetry Status
Shorted Terminals / Fault Bridge	$<0.10\ \Omega$	<500	0x0E --- LOOP FAULT
Good / Armed Igniter Loop	$0.85\ \Omega$	21,242	0x01 --- SYSTEM READY
Degraded Connection Interface	$4.50\ \Omega$	8,415	0x0B --- HIGH RESISTAN
Open Circuit / Severed Nichrome	$>10.0\ \text{k}\Omega$	≈ 0	0x00 --- LOOP OPEN

4.4 Field Testing Results

4.4.1 XBee Range Test

An open-terrain XBee range test was conducted, with the flight-computer transmitter set up at ground level and the base-station receiver walked to intervals of 1 km, 3 km, and 5 km. RSSI vs. distance was recorded via laptop and XCTU. The test confirmed sustained bidirectional packet delivery beyond 5 km with a measured RSSI consistent with the 32.34 dB link-margin prediction. Figure ?? shows the team at the test site. Figure 21 shows the ground-station hardware deployed in the field. Figure 23 shows the ground-station LCD confirming live telemetry reception in the DISARMED state.

4.4.2 Dual Deployment Pop Test

Dual deployment pop tests were conducted in a controlled outdoor area under safety clearance. Both the drogue and main deployment channels were triggered sequentially at 70% PWM duty cycle. Successful ejection was confirmed for both channels without nichrome wire burnout.

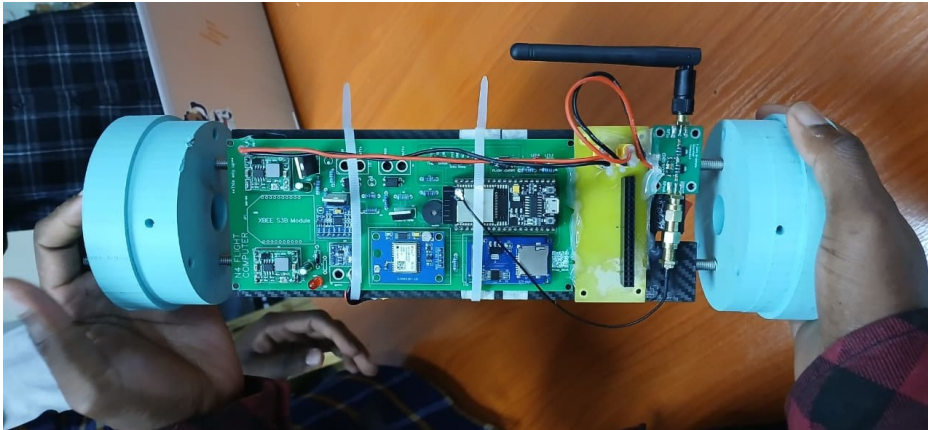


Figure 17: Nakuja recovery team at the field test site during the XBee range test and pop test campaign.

4.4.3 Drone Simulated Flight Test

The fully integrated avionics system was mounted onto a drone platform for a tethered low-altitude simulated flight test. The XBee telemetry link remained active throughout the flight. Apogee detection and deployment sequence logic triggers were confirmed via the FSM state-machine log. All telemetry data were recorded for analysis.

4.5 Firmware Development Progress

4.5.1 Completed Modules

The following firmware modules have been developed and unit-tested:

1. **XBee Communication Module:** Hardware UART driver, binary data framing, and transmission interrupt at 10 Hz.
 2. **GPS Module Interface:** NMEA sentence parsing, coordinate extraction, and satellite fix tracking.
 3. **IMU Interface:** I²C interaction with MPU6050 for real-time acceleration acquisition.
-

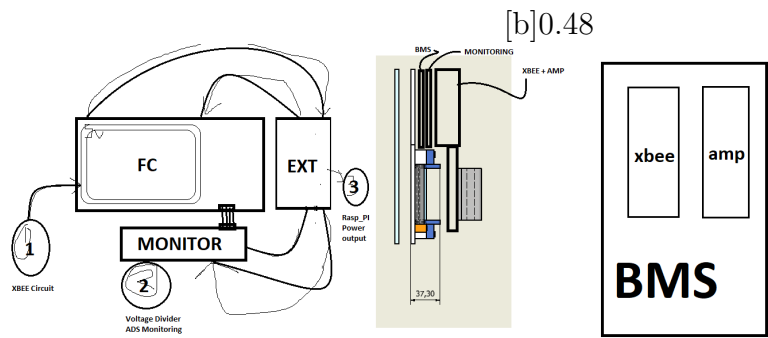
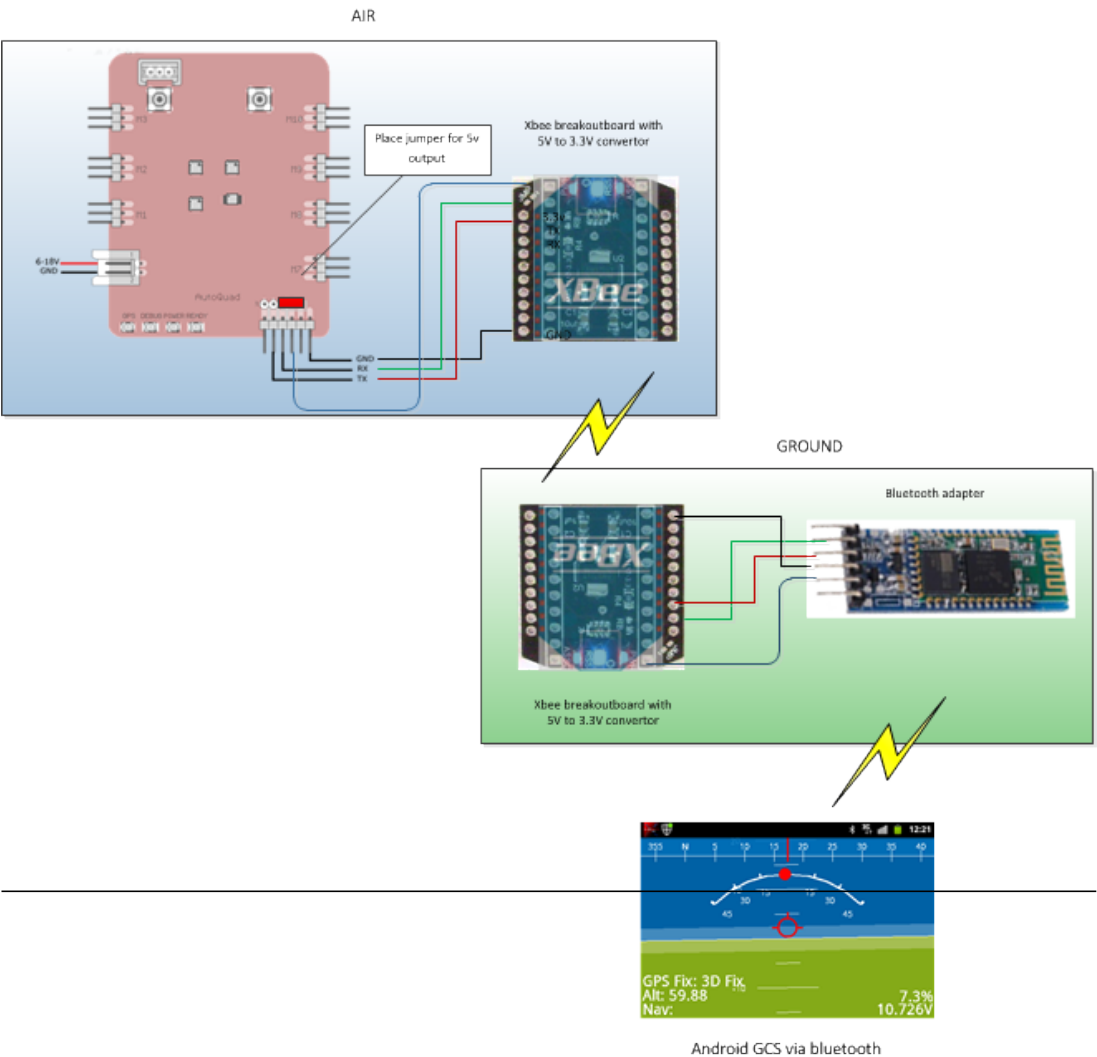


Figure 18: Ground-station electronics and avionics hardware ready for field deployment.

[b]0.48



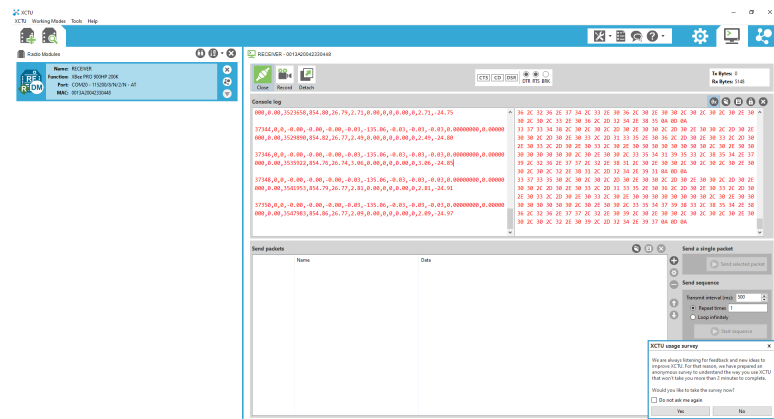


Figure 21: Ground-station LCD readout during range test confirming live telemetry reception from the rocket flight computer over the XBee link, showing battery voltage, RSSI, altitude, and DISARMED state.



Figure 22: Post-pop-test inspection confirming no nichrome burnout and verifying charge-holder structural integrity after dual-deployment sequence.

4. **Barometer Interface:** I²C sampling of BMP280 for temperature-compensated altitude derivation.
5. **Voltage Monitoring:** Moving-average filtering across ADS1115 diagnostic channels.
6. **Continuity Detection:** Real-time terminal resistance calculation and auto-fault tracking.
7. **PWM Control:** Calibrated 1000 Hz gate drive switching at 70% duty cycle to prevent nichrome burnout.

4.5.2 Integration Status

Standalone drivers have been integrated into a concurrent application layer under FreeRTOS thread scheduling, simultaneously: sampling all high-priority sensor buses without clock blocking; transmitting unified telemetry frames via XBee at stable 10 Hz; computing real-time battery status and flagging loop continuity faults; and parsing outbound command instructions from the ground station.

4.5.3 Remaining Firmware Work

- Integrate local flight-logging capabilities on a SPI-based micro-SD card.
- Incorporate dual-sensor altitude fusion (BMP280 + MPU6050) to prevent premature ejection triggers from transonic pressure anomalies.

4.6 Challenges Encountered and Engineering Resolutions

Several technical challenges were isolated and resolved:

- **XBee AT command routing:** Erratic AT command behaviour was resolved by flashing custom parameter profiles directly via XCTU to establish matching network identifiers.
-

- **ESP32 ADC noise:** The native ESP32 ADC experienced excessive wide-band noise when the 2.4 GHz Wi-Fi radio was actively broadcasting. This was mitigated by averaging $N = 10$ continuous data samples.
- **PETG bed adhesion:** Early PETG prints suffered severe thermal contraction warping. This was eliminated by increasing the bed temperature to 80 °C, treating the build plate with specialised adhesives, and lowering print velocity to guarantee dimensional accuracy.

4.7 Discussion

The bench and field testing results confirm that all three primary design objectives have been met. The 32.34 dB link-margin prediction was validated empirically through sustained packet delivery beyond 5 km, with live battery voltage, RSSI, and altitude data confirmed on the ground-station LCD. The PWM ignition strategy at 70% duty cycle eliminated the nichrome burnout problem documented in N-3.5, enabling repeatable dual-deployment pop tests. The ADS1115 voltage reconstruction achieved ± 0.02 V accuracy across the full 12–16.8 V operating window, providing reliable launch-inhibit logic.

The stepped PETG avionics sled successfully survived the 188 kPa pop-test pressure transients with all components intact, validating the distributed fastening and stepped geometry design. The drone flight test demonstrated FSM state sequencing and continuous telemetry under simulated dynamic conditions, bridging the gap between bench validation and actual rocket flight.

These results collectively establish that the N-4 dual-deployment recovery system is ready for full rocket integration and a supervised launch attempt.

5 Conclusion and Recommendations

5.1 Conclusion

This project has demonstrated a fully realised, structurally verified mechatronic solution for the N-4 sounding rocket dual-deployment recovery system, systematically mitigating the legacy constraints of telemetry range limits, pyrotechnic ignition fatigue, and the absence of pre-flight diagnostics.

The integration of the sub-GHz Digi XBee-PRO 900HP transceiver was successfully achieved, verifying a deterministic hardware serial data link. Mathematical link-budget profiling confirmed a robust link margin of 32.34 dB for the 5 km line-of-sight recovery baseline, validated empirically during field range testing with the ground-station LCD confirming live packet reception including battery voltage and RSSI.

Controlled energy delivery across the deployment channels was validated through IRLZ44N MOSFET switching with 70% PWM duty cycle. Dual deployment pop tests confirmed successful drogue and main parachute ejection without nichrome wire structural failure, establishing a reliable alternative to direct-current discharge methods.

Pre-flight systemic feedback was established via the 16-bit ADS1115 ADC over I²C. Empirical bench testing confirmed voltage reconstruction accuracy of ± 0.02 V across the full operating range, enabling automated continuity checks and launch-inhibit safety logic.

The 3D-printed PETG avionics bay sled satisfied all physical constraints within the 95 mm inner airframe diameter. The stepped battery seating geometry bypassed internal threaded structural rod trajectories while providing vibration isolation and secure battery retention under launch and pop-test loading.

The embedded firmware was deployed as a concurrent multi-threaded architecture under FreeRTOS on the ESP32 microcontroller. A drone-mounted simulated

flight test confirmed active telemetry and correct FSM state sequencing under dynamic conditions. The N-4 mechatronic recovery suite provides a scalable and robust design methodology for student-led aerospace initiatives.

5.2 Recommendations for Future Work

1. **Non-Volatile Black-Box Data Logging:** Integrate a high-speed SPI micro-SD card logging module within the FreeRTOS framework to provide full onboard data redundancy against transient telemetry dropouts at apogee.
 2. **Advanced Sensor Fusion for Apogee Detection:** Incorporate a dual-sensor filtering matrix (BMP280 barometer + MPU6050 accelerometer) within the FSM logic to prevent premature ejection triggers caused by transonic pressure anomalies.
 3. **Bi-Quad Base-Station Antenna Upgrade:** Replace the duck dipole ground-station antenna with a directional Bi-Quad at 900 MHz to improve null-zone robustness and extend effective link range during descent phases when the rocket antenna axis may align with the ground-station direction.
 4. **Full Rocket Integration and Launch:** Complete all hardware integration, conduct a final end-to-end rehearsal test, and proceed to a supervised launch at the Isiolo School of Infantry Range as an entry into the Spaceport America Cup.
 5. **Parachute Sizing and Drop Test:** Validate drogue and main parachute sizing against the N-4 target descent rate profile using a drop test from a tethered balloon or elevated platform.
-

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A CAD Drawings with Title Blocks

The engineering drawings below were produced in SolidWorks and include the institution title block, part number, material specification, and revision level required for fabrication. All dimensions are in millimetres.

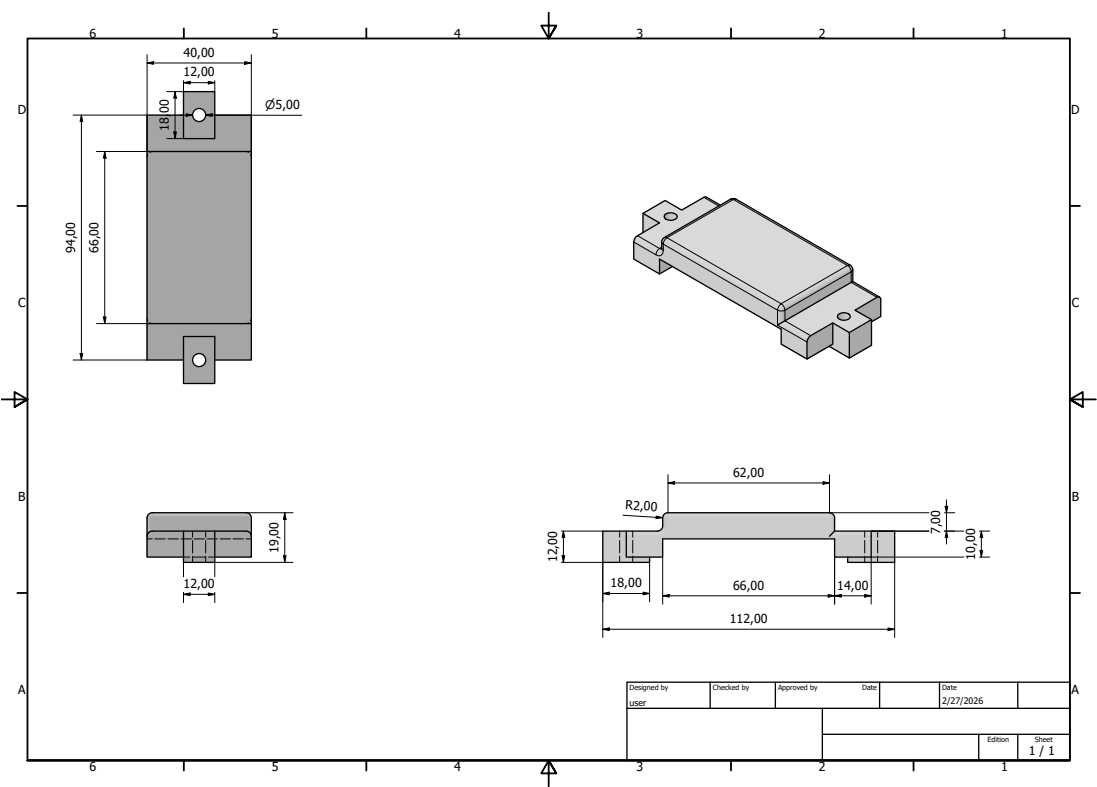


Figure 23: 18650 battery case cover – 2D manufacturing drawing with title block

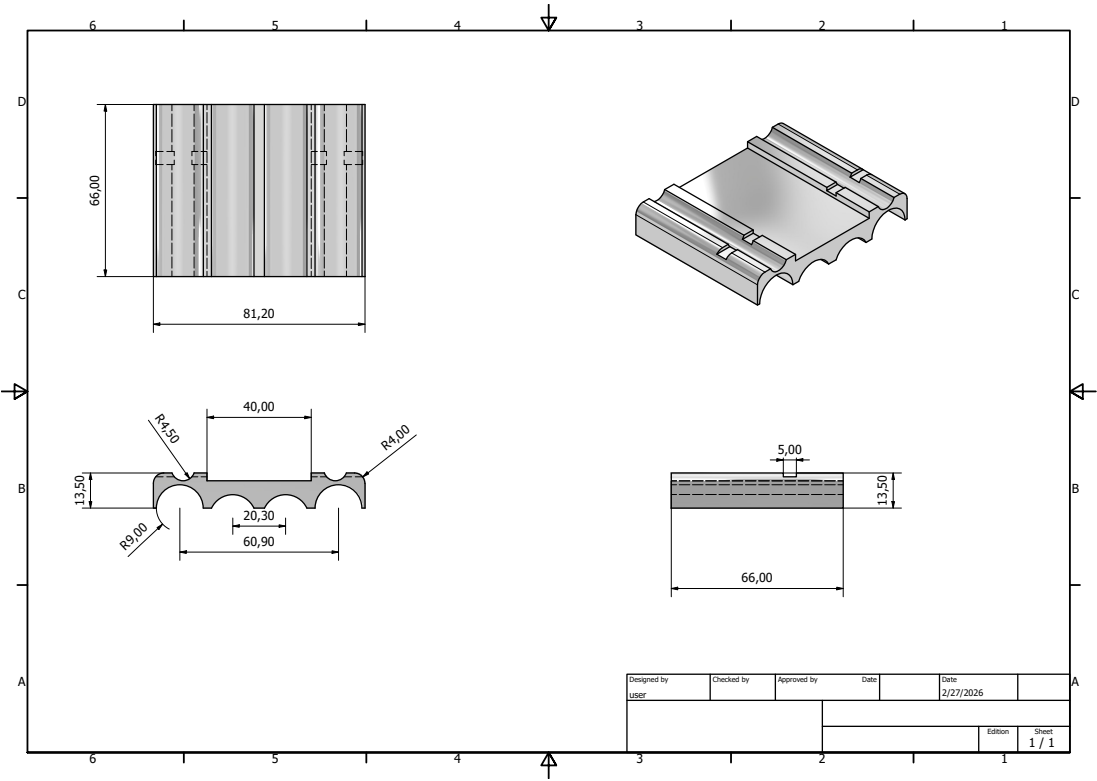


Figure 24: 18650 battery case body – 2D manufacturing drawing with title block

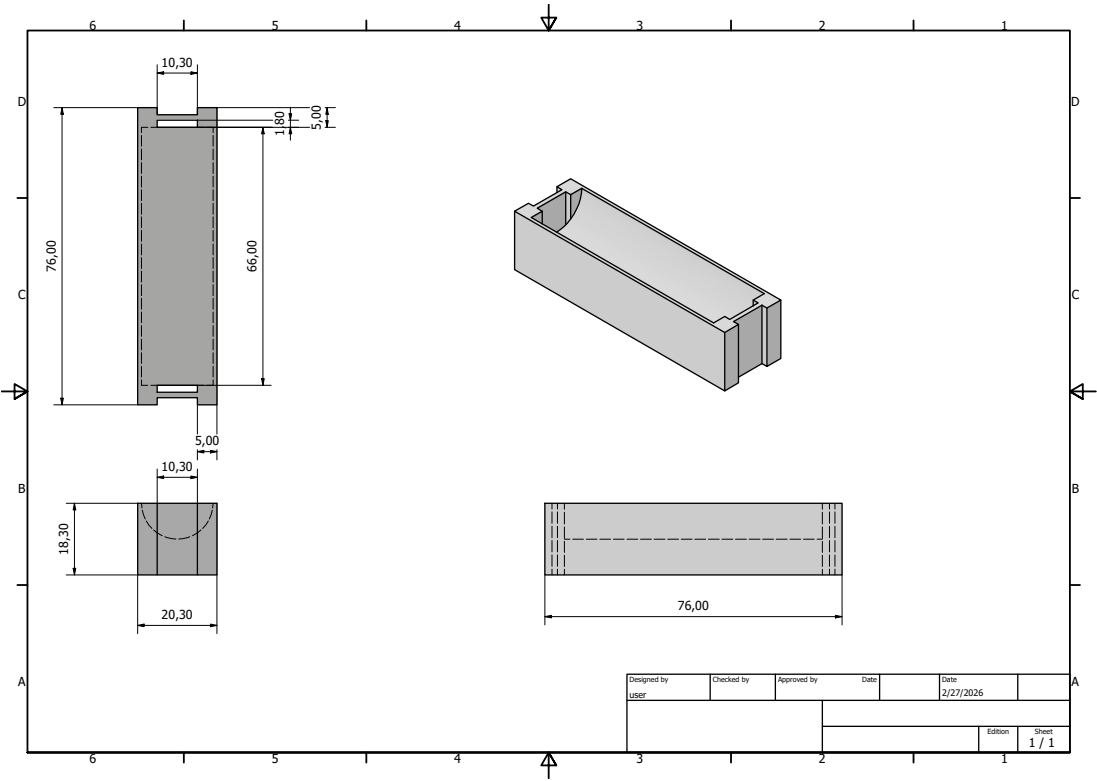


Figure 25: Edge battery support bracket – 2D manufacturing drawing with title block

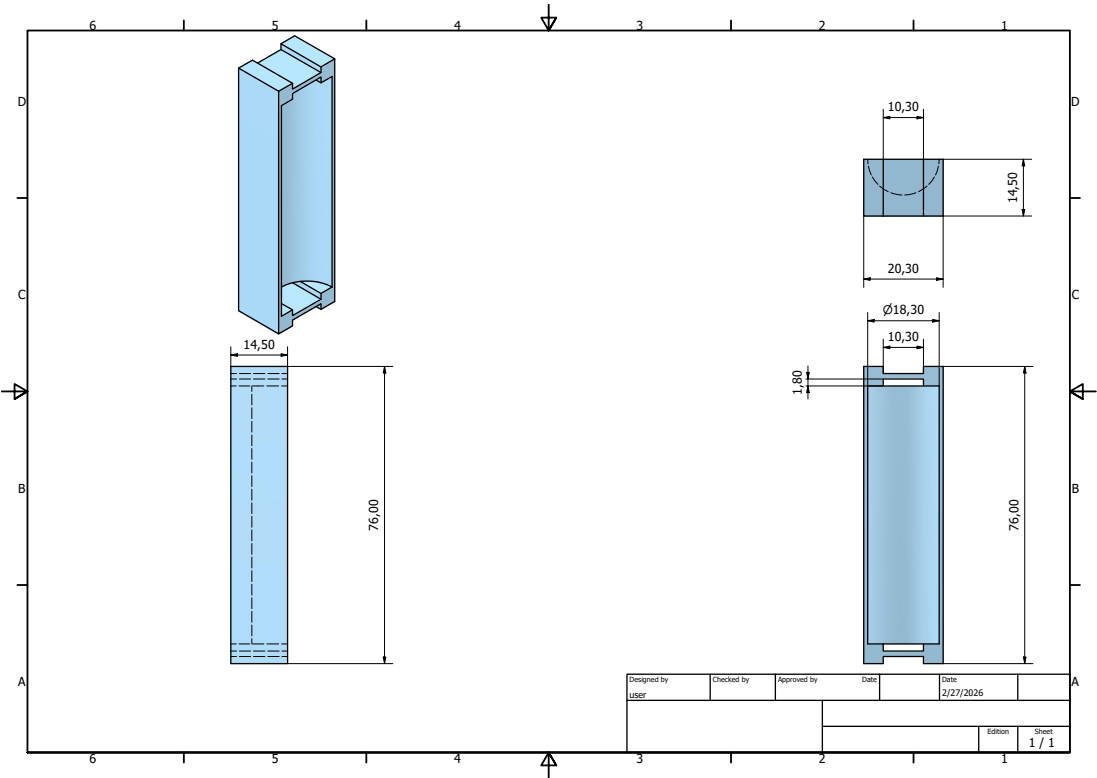


Figure 26: Mid battery support bracket – 2D manufacturing drawing with title block

B Flight Computer and Base Station Circuit Design Reference

Figures 39 and 40 present the complete EasyEDA schematics used as design references for PCB layout, component placement, and trace routing.

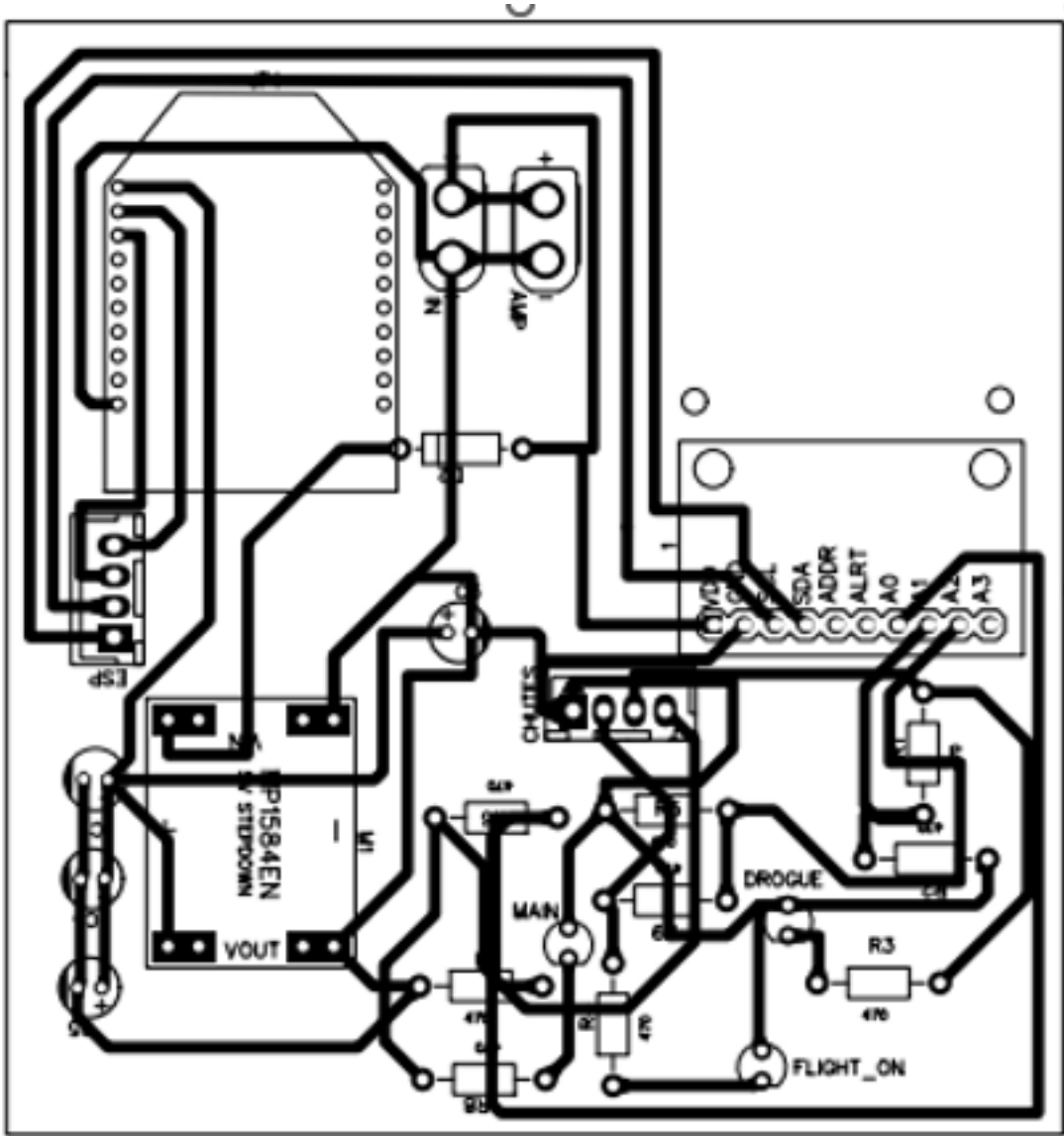


Figure 27: Flight computer EasyEDA schematic with ESP32, XBee, MOSFETs, and ADS1115 monitoring circuit.

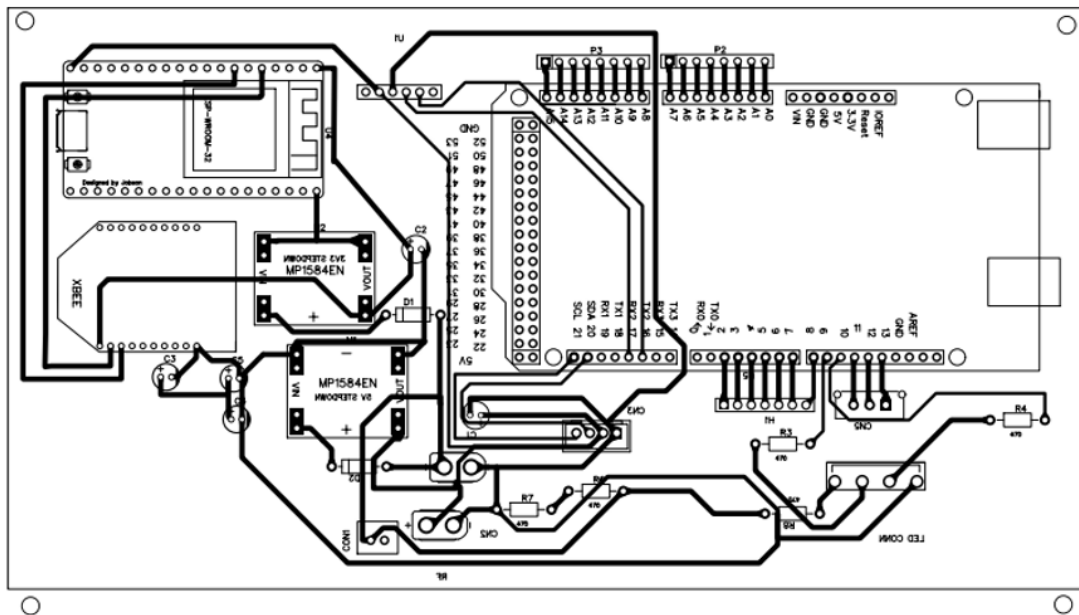


Figure 28: Base station EasyEDA schematic with XBee UART and Bluetooth interface.

C Project Budget

Table 18: Project Budget Breakdown

Item	Qty	Unit Cost (KES)	Total (KES)	Status
ESP32-WROOM-32	2	1,800	3,600	Received
XBee-PRO 900HP	2	4,500	9,000	Received
4S Lithium-ion batteries (30,000 mAh)	4	500	2,000	Received
BMS Board (4S)	3	650	1,950	Received
IRLZ44N MOSFET	10	45	450	Received
Nichrome Wire (10 m)	1	500	500	Received
PCB Fabrication (5 pcs)	1	4,500	4,500	Fabricated
PCB Components (R, C, etc.)	1	2,000	2,000	Partial
PETG Filament (1 kg)	1	1,800	1,800	Received
Crimson Powder (100 g)	1	500	500	Pending
Connectors (XT60, JST, etc.)	1	1,200	1,200	Received
Fasteners and Hardware	1	800	800	Partial
Miscellaneous	1	1,500	1,500	—
Total Estimated Cost:			29,800	—

D Production Plan

Table D presents the Semester 2 production plan for the N-4 recovery system, covering all fabrication, integration, testing, and documentation activities from March 2026 to June 2026.

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p1cmp4.5cmp3cmp2.5cmp2.5cmp1.5cm Production Plan — Final Year Project
2025/2026 (Semester 2: March–June 2026)

Wk	Tasks / Activities	Materials Required	Special Equipment	Sequential / Concurrent	Status
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(Table D continued)

Wk	Tasks / Activities	Materials Required	Special Equipment	Sequential / Concurrent	Status
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Continued on next page

1 1. FC Circuit Update: Review FC PCB; source replacement components; re-etch board; re-solder and retest. **2. Software — Battery/Chute Monitoring:** Merge ADS1115 monitoring FreeRTOS task into main FC firmware; confirm 14.9 V nominal detection; confirm launch-inhibit triggers at <14.0 V. PCB copper clad; HCl + H₂O₂; solder, flux Laptop + serial monitor; soldering iron; multimeter Circuit update concurrent with SW integration Done

2 1. 3D Printing — FC Sled Holder: Slice in Prusa Slicer (PETG, 0.2 mm layer, 40% infill); print; test-fit with FC PCB and brass column mounts. **2. Base Station Circuit — Soldering:** Transfer schematic to perfboard; solder all components; test power rails and XBee UART. PETG filament; perfboard; solder, flux; header pins Prusa 3D printer; Prusa Slicer; callipers; soldering iron; multimeter; laptop FC sled print after printer available; base station soldering concurrent Done

3 1. Base Station 3D Print and Housing. 2. Ejection Caps Reprint: Revised geometry; test-fit nichrome wire and seal. **3. Dual Deployment Pop**

Test: Safety clearance; load caps; trigger drogue at 70% PWM; trigger main at 70% PWM; confirm sequence; iterate if needed. PETG filament; perfboard; solder; black powder; nichrome wire; all-weather container Prusa 3D printer; soldering iron; multimeter; laptop; safety PPE; fire extinguisher Sled print + BS solder concurrent; pop test after cap fit confirmed Done

4 1. Full Avionics Integration — Bench Test: Mount FC PCB into PETG sled; connect battery, XBee, and deployment circuit; run FreeRTOS firmware; confirm launch inhibit, continuity detection, and fire logic. **2. Base Station Final Assembly.** 4S 18650 battery pack; full avionics assembly Laptop + serial monitor; multimeter; soldering iron Bench integration after FC sled + SW done; BS assembly concurrent Done

5 Dual Deployment Pop Test (repeat): Safety clearance; trigger drogue at 70% PWM; trigger main at 70% PWM; confirm sequence. 4S 18650 battery pack; full avionics assembly Laptop + serial monitor; multimeter; soldering iron BS assembly concurrent with bench integration Done

6 Dual Deployment Pop Test (verification): Confirm drogue and main fire at 70% PWM; record results. 4S 18650 battery pack; full avionics assembly Laptop + serial monitor; multimeter; soldering iron After all bench integration passes Done

7 1. XBee Range Test (Field): Set up FC transmitter at ground level; walk base station to 1 km, 3 km, 5 km; record RSSI vs. distance via XCTU; validate ≥ 5 km link margin. **2. System Integration Refinement. 3. Report & Documentation Update.** Both XBee modules; laptops; transport Laptop + XCTU software Range test after full bench integration; documentation concur-

rent Done

8 **1. Drone Test (Simulated Flight):** Mount integrated avionics onto drone; tethered low-altitude flight; verify XBee telemetry active during flight; confirm apogee detection and deployment sequence logic; record all telemetry. **2. Pre-Launch System Validation.** Full avionics assembly; drone platform; tether line Laptop + serial monitor; multimeter; drone Drone test after Wk 7 refinement; system validation concurrent Incomplete

9 **1. Demonstration:** Prepare demonstration script and safety checklist; conduct supervised ground demonstration; document results. **2. Final Report:** Write Chapter 4 (fabrication, test results, discussion); write Chapter 5 (conclusions); compile appendices; supervisor review and submission. **3. Final Presentation:** Prepare and rehearse final slide deck; deliver to panel. Full system assembly; test fixtures Laptop; camera/recorder Demonstration after system validation sign-off; final report concurrent; presentation concurrent with report Pending

E Labelled Assembly Reference Photographs

The labelled images below serve as fabrication and assembly references, identifying all physical components and their spatial relationships.

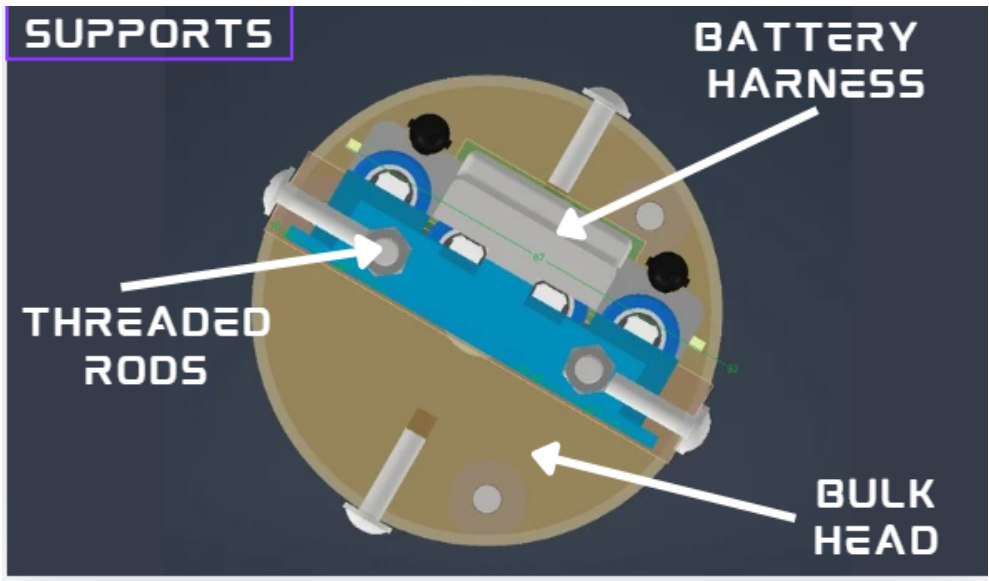


Figure 29: Labelled cross-sectional avionics bay with component stacking and clearances

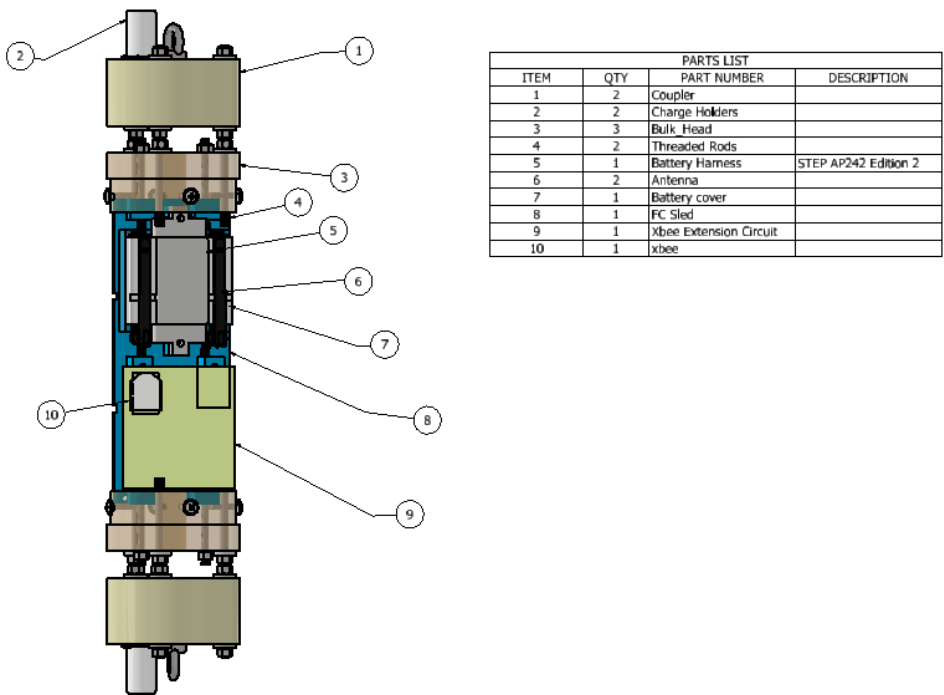


Figure 30: Labelled assembly parts list with component identifications

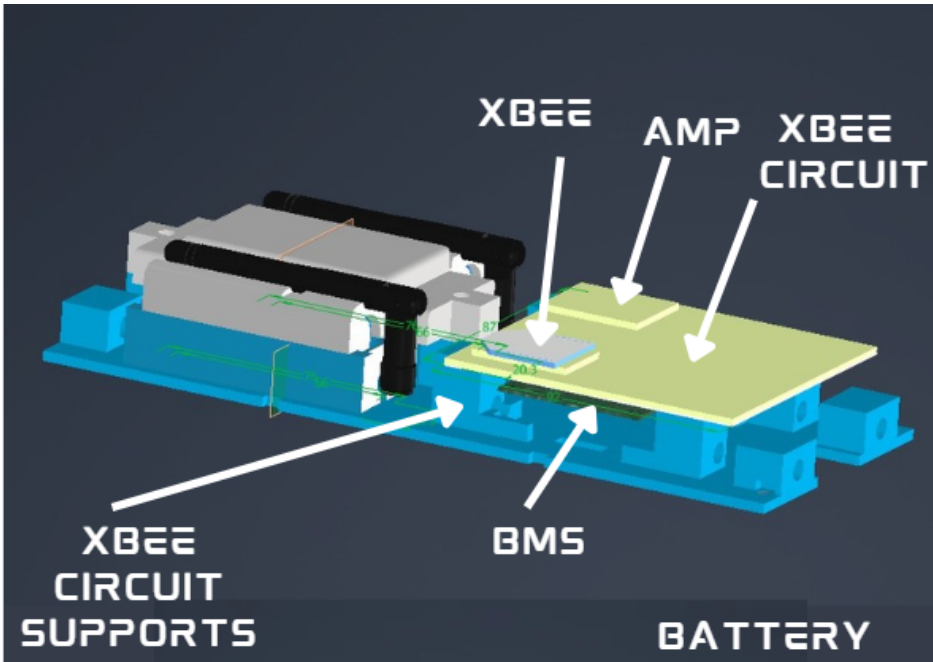


Figure 31: Labelled avionics bay showing battery pack, sled, and connectors

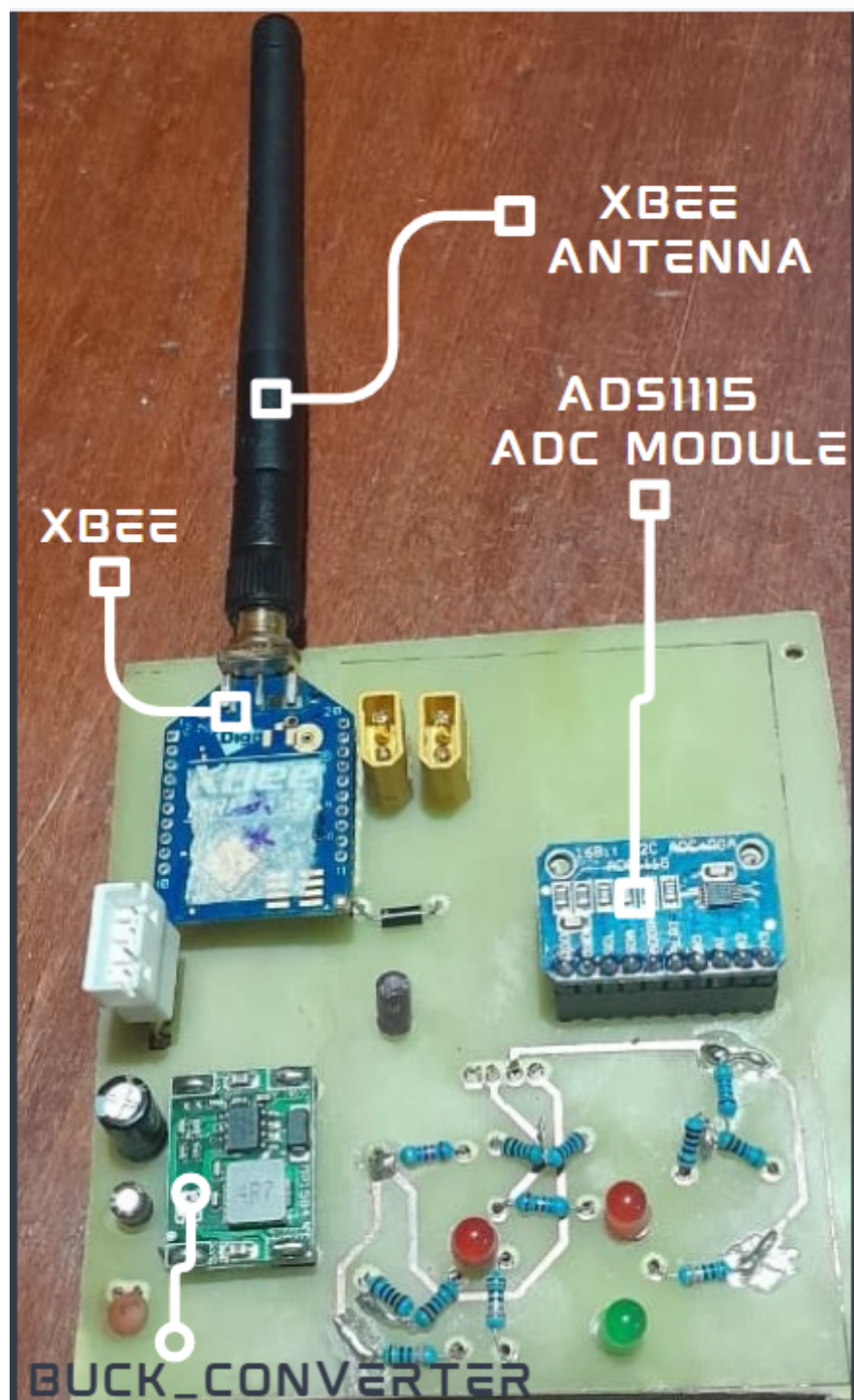


Figure 32: Labelled flight-computer PCB with integrated electronic components

F Firmware Code Snippets and Repository

The N-4 flight software is version-controlled and publicly available at:

<https://github.com/nakujaproject/N4-Flight-Software> [37]

F.1 XBee UART Initialisation Task

```
1 void xbee_init_task(void *pvParameters) {
2     uart_config_t uart_cfg = {
3         .baud_rate    = 115200,
4         .data_bits    = UART_DATA_8_BITS,
5         .parity       = UART_PARITY_DISABLE,
6         .stop_bits    = UART_STOP_BITS_1,
7         .flow_ctrl    = UART_HW_FLOWCTRL_DISABLE,
8     };
9     uart_param_config(UART_NUM_2, &uart_cfg);
10    uart_set_pin(UART_NUM_2, TX2_PIN, RX2_PIN,
11                UART_PIN_NO_CHANGE, UART_PIN_NO_CHANGE);
12    uart_driver_install(UART_NUM_2, BUF_SIZE, 0, 0, NULL, 0);
13
14    /* Enter AT command mode and set destination address */
15    xbee_send_at("+++");          vTaskDelay(pdMS_TO_TICKS(1200))
16    ;
17    xbee_send_at("ATDH_0013A200"); vTaskDelay(pdMS_TO_TICKS(100))
18    ;
19    xbee_send_at("ATDL_41B5C2F3"); vTaskDelay(pdMS_TO_TICKS(100))
20    ;
21    xbee_send_at("ATWR");          vTaskDelay(pdMS_TO_TICKS(200));
22    xbee_send_at("ATCN");          /* Exit command mode */
23    vTaskDelete(NULL);
24 }
```

Listing 1: XBee UART initialisation and AT-command configuration (excerpt from `xbee_task.c`) [37]

F.2 ADS1115 Battery Monitoring Task

```

1 void power_monitor_task(void *pvParameters) {
2     ads1115_t ads = ads1115_init(I2C_NUM_0, ADS1115_ADDR_GND);
3     while (1) {
4         int16_t raw = ads1115_read_channel(&ads, ADS1115_MUX_AIN0
5             );
6         /* Reconstruct: Vbus = raw * (4.096/32768) * (R1+R2)/R2
7            */
8         float v_sense = raw * (4.096f / 32768.0f);
9         float v_bus    = v_sense * ((47000.0f + 10000.0f) /
10             10000.0f);
11
12         if (v_bus < VBUS_MIN_V) {
13             launch_inhibit_set(INHIBIT_LOW_BATT);
14         } else {
15             launch_inhibit_clear(INHIBIT_LOW_BATT);
16         }
17         telemetry_packet.battery_mv = (uint16_t)(v_bus * 1000.0f)
18             ;
19         vTaskDelay(pdMS_TO_TICKS(500));
20     }
21 }

```

Listing 2: ADS1115 I²C voltage monitoring and launch-inhibit logic (excerpt from power_monitor.c) [37]

F.3 PWM Deployment Task

```

1 #define LEDC_CHANNEL_DROGUE LEDC_CHANNEL_0
2 #define LEDC_CHANNEL_MAIN   LEDC_CHANNEL_1
3 #define PWM_FREQ_HZ         1000
4 #define PWM_DUTY_PCT        70    /* 70% validated duty cycle */
5

```

```

6 void pyro_task(void *pvParameters) {
7     ledc_timer_config_t timer = {
8         .speed_mode      = LEDC_LOW_SPEED_MODE,
9         .duty_resolution = LEDC_TIMER_8_BIT,
10        .timer_num       = LEDC_TIMER_0,
11        .freq_hz         = PWM_FREQ_HZ,
12    };
13    ledc_timer_config(&timer);
14    uint32_t duty = (PWM_DUTY_PCT * 255) / 100; /* = 178 */
15    while (1) {
16        EventBits_t ev = xEventGroupWaitBits(deploy_event_group,
17            DEPLOY_DROGUE_BIT | DEPLOY_MAIN_BIT,
18            pdFALSE, pdFALSE, portMAX_DELAY);
19        if (ev & DEPLOY_DROGUE_BIT)
20            ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
21                                    LEDC_CHANNEL_DROGUE, duty,
22                                    0);
23        if (ev & DEPLOY_MAIN_BIT)
24            ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
25                                    LEDC_CHANNEL_MAIN, duty, 0);
26        vTaskDelay(pdMS_TO_TICKS(3000)); /* fire window */
27        ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
28                                LEDC_CHANNEL_DROGUE, 0, 0);
29        ledc_set_duty_and_update(LEDC_LOW_SPEED_MODE,
30                                LEDC_CHANNEL_MAIN, 0, 0);
31    }
32 }

```

Listing 3: IRLZ44N MOSFET PWM deployment channel control (excerpt from `pyro_task.c`) [37]

F.4 Finite-State Machine Transition Logic

```

1 typedef enum {

```

```
2     STATE_PAD, STATE_ARMED, STATE_BOOST,
3     STATE_APOGEE, STATE_DROGUE, STATE_MAIN, STATE_LANDED
4 } flight_state_t;
5
6 flight_state_t fsm_update(flight_state_t current, sensor_data_t *
7     s) {
8     switch (current) {
9         case STATE_PAD:
10             return (s->armed_flag) ? STATE_ARMED : STATE_PAD;
11         case STATE_ARMED:
12             return (s->accel_z > LAUNCH_THRESH_G) ? STATE_BOOST
13                                                         : STATE_ARMED;
14         case STATE_BOOST:
15             return (s->accel_z < BURNOUT_THRESH_G) ? STATE_APOGEE
16                                                         : STATE_BOOST
17                                                         ;
18         case STATE_APOGEE:
19             xEventGroupSetBits(deploy_event_group,
20                 DEPLOY_DROGUE_BIT);
21             return STATE_DROGUE;
22         case STATE_DROGUE:
23             return (s->altitude_m < MAIN_DEPLOY_ALT_M) ?
24                 STATE_MAIN
25                 :
26                 STATE_DROGUE
27                 ;
28         case STATE_MAIN:
29             xEventGroupSetBits(deploy_event_group,
30                 DEPLOY_MAIN_BIT);
31             return STATE_LANDED;
32         default: return STATE_LANDED;
33     }
34 }
```

Listing 4: FSM state transition logic for flight phase detection (excerpt from `state_machine.c`) [37]

F.5 Supporting Figures and Diagrams

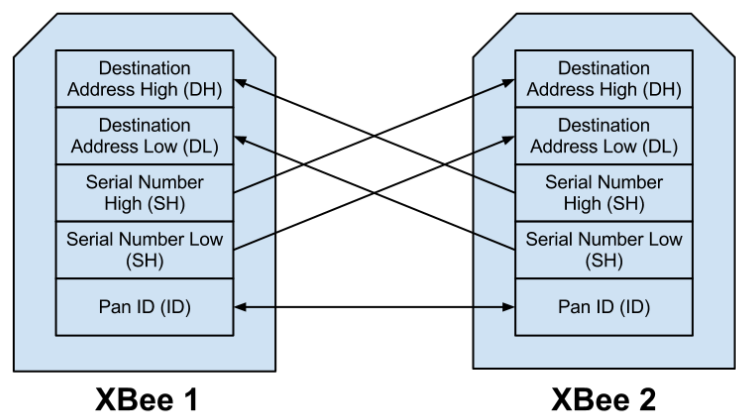


Figure 33: N-4 XBee telemetry architecture for rocket-to-ground communication [25, 26]



Figure 34: 900 MHz duck antenna used in the N-4 telemetry configuration

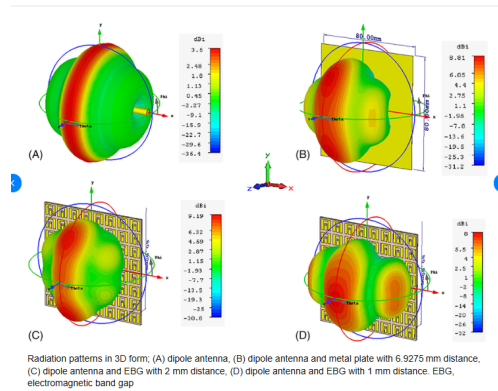


Figure 35: Dipole radiation “donut” pattern illustrating nulls along the antenna axis

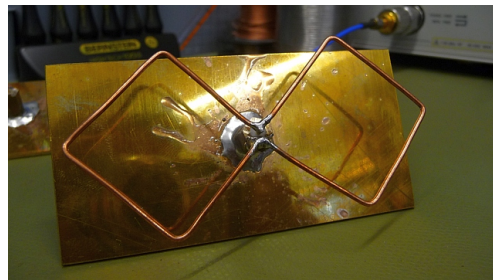


Figure 36: 900 MHz Bi-Quad ground-station antenna option

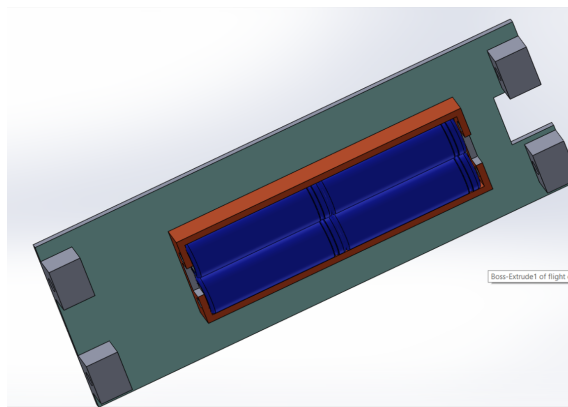


Figure 37: Old flight-computer holder failure after pop tests

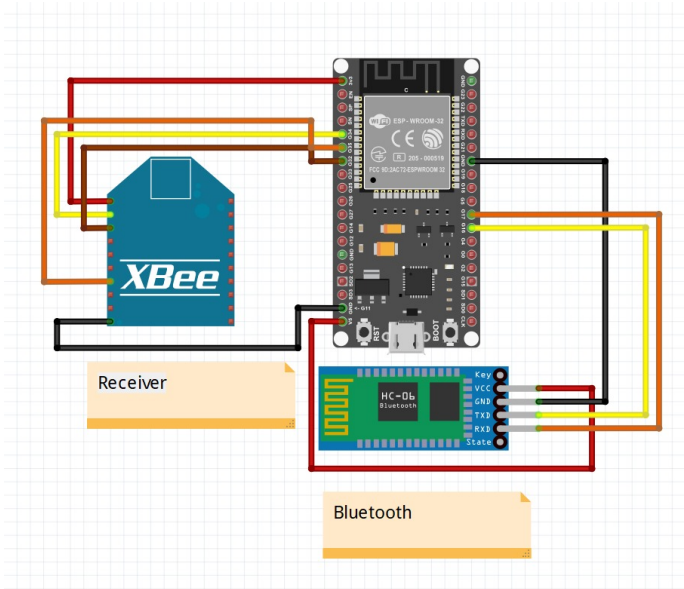


Figure 38: Base-station interconnect layout (Fritzing schematic)

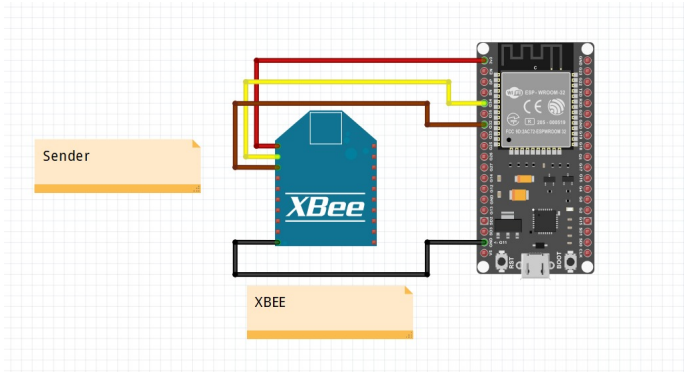


Figure 39: Rocket XBee extension wiring layout (Fritzing schematic)

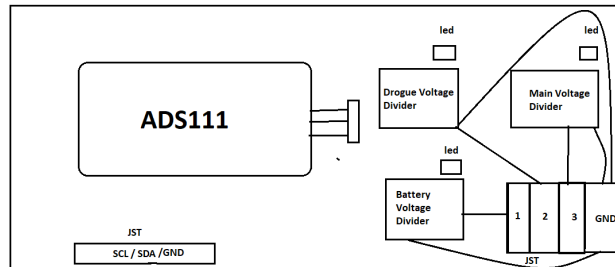


Figure 40: Power monitoring hardware concept with ADS1115 divider and launch inhibit path

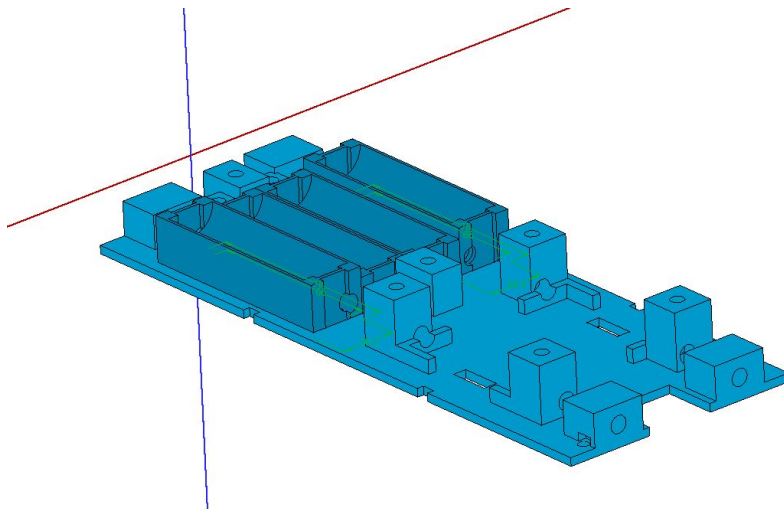


Figure 41: Flight-computer holder sled 3D CAD model

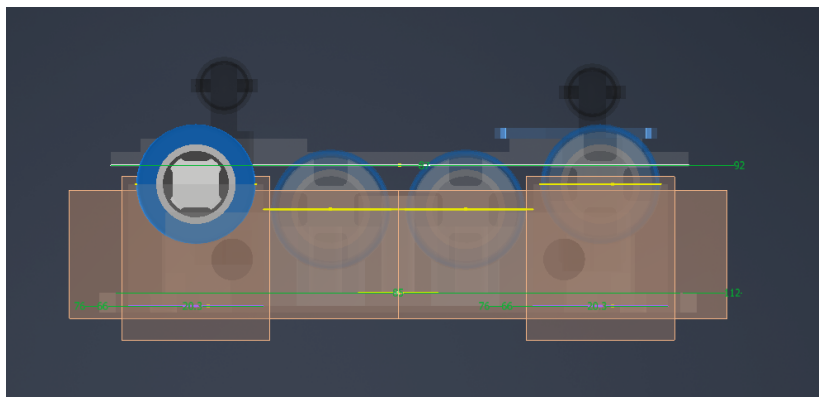


Figure 42: Threaded-rod integration showing raised battery cell clearance

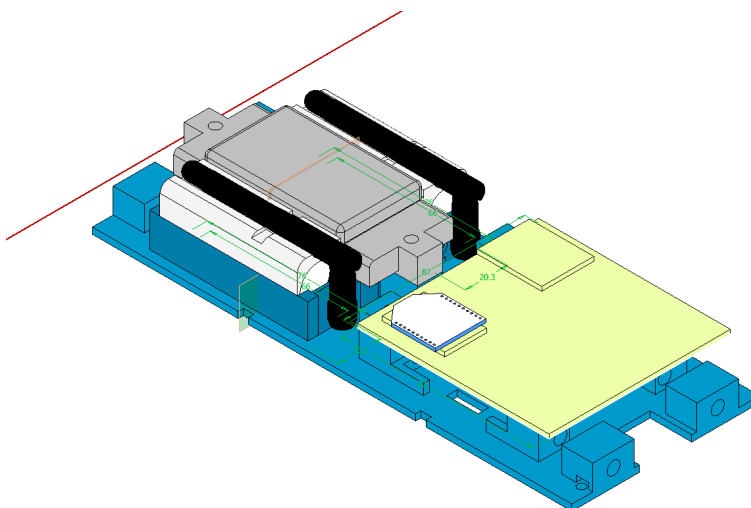


Figure 43: Avionics bay assembly showing battery, holder, and electronics integration